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How does blockchain mitigate false advertising in live streaming E-commerce? A tripartite stochastic evolutionary game approach

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ABSTRACT

The rapid rise of live-streaming e-commerce, particularly in China, has transformed consumer shopping experiences; however, it has also introduced challenges such as false advertising and distorted product information. This study explores how blockchain technology can address these issues within the live-streaming e-commerce ecosystem. Using a tripartite evolutionary game model involving e-commerce platforms, merchants, and consumers, we apply stochastic differential equations to analyze the stability of strategic behaviors and dynamics of false advertising. Key findings include: (1) In high-commission environments, platforms and merchants are better aligned, increasing the likelihood of merchants adopting truthful advertising strategies. (2) In high-profit environments, the temptation for merchants to engage in false advertising rises, prompting platforms to adopt blockchain technology to maintain transparency. (3) Although blockchain technology adoption enhances platform reputation over the long term, its direct impact on merchants' advertising strategies is limited. (4) Regulatory measures such as fines and consumer reporting rewards promote blockchain technology adoption and reduce false advertising. These findings provide valuable insights for enhancing the credibility of live-streaming e-commerce and offer guidance for platforms on balancing technology adoption with regulatory efforts to foster sustainable industry development.

1. Introduction

"Live streaming + e-commerce," an emerging sales model, has gained significant momentum globally (Xin et al., 2023). According to Peng et al. (2024), the global video streaming market totaled USD 106.83 billion in 2023 and is projected to grow at a compound annual rate of 21.5 % from 2023 to 2030, highlighting its tremendous growth potential. Among the most notable markets is China, where live-streaming e-commerce has experienced explosive growth. According to Statista (2023), the live streaming e-commerce market in China totaled nearly 5 trillion yuan in 2023, a significant increase from 420 billion yuan in 2019, and is projected to surge to 8.16 trillion yuan by 2026. Beyond China, live-streaming e-commerce has also gained traction in regions such as the United States and Europe (Aastha, 2023; Hu and Chaudhry, 2020).

The rise of e-commerce has greatly enriched consumers' access to goods, but has also widened the spatial distance between consumers and retailers, providing opportunities for false advertising. The Economic Times (2023)reveals that in India, the compliance rate for digital advertising is only 75 %, with 81 % of violations involving misleading claims.

In a traditional e-commerce environment, many goods including food and pharmaceuticals are often misrepresented by merchants who exaggerate their functions or effects (Aikin et al., 2024; Lachenmeier et al., 2013; Schwartz and Woloshin, 2019). For instance, when advertising weight loss products, some businesses may exaggerate the effectiveness of their products by manipulating small sample studies or graphical presentations. Advertisers may collaborate with profit-biased researchers to conduct studies with only a small number of subjects, claiming up to 50 % weight loss while omitting the extremely small sample size of the study, thereby misleading consumers about the product (Single Grain, 2023). In addition, businesses may further mislead consumers' purchasing behavior by manipulating online reviews and using fake or paid reviews to exaggerate product quality (Poddar et al., 2019; Zheng et al., 2013).

Compared with the traditional text-based e-commerce model, the live-streaming one provides consumers with a more immersive and dynamic shopping experience (Luo et al., 2021). Products are no longer

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showcased solely through beautiful images, detailed descriptions, and customer reviews; instead, language becomes a key element (Zhang S. et al., 2022). Through vivid language expressions by streamers, products are presented in a more captivating way, enabling real-time viewers to gain a more comprehensive perception and experience of the product. In this model, language is given a special role as a powerful and emotionally resonant tool (Li et al., 2022). Carefully crafted words can quickly evoke emotional responses in consumers, build connections, inspire shopping desires, and enhance the joy and engagement of the shopping experience.

However, the power of language in this context can also be abused, rendering it a new tool for false advertising (Hong, 2019). Beyond misleading descriptions and exaggerated slogans by streamers, deceptive practices often involve background boards, cue cards, subtitles, and other visual elements, further distorting the presentation of product information. For example, the widely publicized TikTok merchant Oriental Selection scandal involved a streamer repeatedly misrepresenting "farmed shrimp" as "wild shrimp," misleading consumers through exaggerated claims supported by background cue cards, images, and coordinated voiceovers from the team. Another case is the controversy surrounding Taobao streamer Li Jiaqi, who used exaggerated urgency to promote his products by declaring, "the last item in the gift pack—buy it now." Streamers' false advertising can drive impulsive buying behavior, resulting in buyer's remorse, increased return rates, and disruptions to the normal trading order in live e-commerce (Kumar and Taneja, 2024; Ming et al., 2021; Wu et al., 2023).

Efforts to combat false advertising in e-commerce rely on government regulations, legal measures, and technological innovations. For instance, the Chinese Consumer Protection Law imposes strict penalties for deceptive advertising (French, 2008). However, legal enforcement faces challenges including ambiguous liability determination, difficulties in collecting evidence, and inconsistent consumer attitudes. Similarly, while artificial intelligence technologies have been introduced to monitor and detect false advertisements, their development in this field remains insufficient (Karode and Werapun, 2022).

Blockchain, a distributed ledger technology, is distinguished by its decentralization and immutability (Alzahrani and Aljabri, 2023; Cong and He, 2019), making it a potential tool to mitigate exaggerated claims about product performance by streamers and provide consumers with more reliable and accurate product information (Liu et al., 2024). Research focuses on the technological applications of blockchain in improving live e-commerce operations and advancing stakeholder interests. However, critical challenges—high costs, complexity, and privacy concerns—remain regarding stakeholders' incentives to adopt blockchain technology and its practical feasibility in addressing information asymmetry. Accordingly, this study explores the following research questions:

(1) How can blockchain technology effectively reduce false advertising in live-streaming e-commerce?
(2) What key factors influence stakeholders' adoption of blockchain technology?
(3) How do different incentives impact the behavior of platforms, merchants, and consumers?

To address these questions, this study employs stochastic evolutionary game theory and blockchain technology analysis to examine the tripartite interactions among platforms, merchants, and consumers. It makes key contributions to the literature on live-streaming e-commerce and blockchain technology. First, it provides insights into how high commission environments can align platform and merchant interests, encouraging the adoption of truthful advertising strategies. Second, it explores how blockchain technology can help platforms maintain transparency and counteract merchants' temptation to engage in false advertising, particularly in high-profit environments. Third, it emphasizes the importance of regulatory measures such as fines and consumer reporting rewards in enhancing blockchain adoption and reducing false advertising, offering practical guidance for fostering industry credibility and long-term sustainability.

2. Literature review

2.1. Blockchain technology

2.1.1. Blockchain technology: definition, origins, and applications

Blockchain technology is a decentralized distributed database that facilitates secure data storage and transmission through the integration of mathematical algorithms, cryptographic protocols, and computer science technologies. By leveraging timestamping services, consensus mechanisms, and asymmetric encryption, blockchain technology eliminates the need for intermediary institutions to establish trust (Wu et al., 2022). The term "blockchain" emerged with the introduction of Bitcoin by Satoshi Nakamoto in 2008 (Nakamoto, 2008). Subsequent research has demonstrated the theoretical potential of the smart contract and digital asset functionalities of blockchain technology to ensure decentralization, transparency, and traceability (Treiblmaier and Sillaber, 2021). Notably, Ethereum's adoption of smart contract technology has been pivotal in its success. These contracts, based on an "if-then" logic, automatically execute and update the blockchain technology upon meeting predefined conditions, ensuring transaction immutability and operational efficiency (Alladi et al., 2019; Pu et al., 2017).

The application of blockchain technology has expanded to various sectors including food safety, finance, supply chain management, and healthcare (Kuo et al., 2017; Li, D. et al., 2024; Rejeb et al., 2020; Wang et al., 2023). For example, Tencent has introduced a blockchain technology-powered electronic contract applet to enhance transactional security (Mercy Adeola, 2024). Similarly, Alibaba Cloud (2024) has integrated blockchain technology with its cloud services to support cross-enterprise and cross-regional collaborations.

2.1.2. Blockchain technology: advantages

Blockchain technology offers several advantages including decentralization, trustlessness, transparency, immutability, and anonymity (Sharma et al., 2017). A blockchain technology network has no central control node or point of concentrated power; all nodes are equal in terms of rights and responsibilities (Yuan and Wang, 2018). During the information recording process, transaction data is replicated and confirmed by multiple nodes (Yaqoob et al., 2022). Once external data is entered into the blockchain technology network via consensus algorithms, the newly generated blocks are updated in real-time across the entire network, with all nodes collectively maintaining the system (Jiang and Chen, 2021; Maesa and Mori, 2020). These characteristics provide a solid technological foundation for the widespread adoption of blockchain technology in various industries. Studies indicate that consumers exhibit a stronger preference for products empowered by blockchain technology. For example, a study on blockchain technology traceability in Chinese tea found that compared to traditional certification methods, consumers were more inclined to choose tea with this traceability (Rao et al., 2023). Violino et al. (2019) investigated Italian consumer preferences for extra virgin olive oil with blockchain-based traceability platforms, revealing that 45 % of consumers opted for QR codes protected by the blockchain technology system (QR-B) and were willing to pay a premium of 17.8 %. Furthermore, regarding Chinese consumers' demand for US beef, Lin et al. (2022) found that approximately 37 % were willing to pay a premium for beef that could be traced through blockchain technology.

2.1.3. Smart contract in e-commerce

In the e-commerce environment, it is more meaningful to proactively prevent issues using smart contracts than addressing them retrospectively (De Graaf, 2019). Scholars have investigated and proposed solutions for key aspects of e-commerce, such as transaction payments,

product circulation, and reputation management. For instance, Ismanto et al. (2019) highlighted that blockchain technology and its smart contract platforms can effectively address issues like fraud, opaque commission structures, limited interactions between buyers and sellers, and misuse of personal data in Indonesian e-commerce platforms. Some studies explored the transformative potential of blockchain technology and crowdsourcing in the e-commerce logistics industry, expanding the application scenarios of smart contracts (Nadime et al., 2023). Moreover, Pranto et al. (2022) and Doğan and Hacer (2023) designed and refined smart contract-based mechanisms including seller reputation systems and authentic review maintenance, enhancing the credibility and fairness of transactions. Wang et al. (2019) proposed a smart contract-supported distributor-to-consumer model that enables real-time monitoring of products and carriers in the e-commerce logistics process, reducing the circulation of counterfeit goods. These studies demonstrate the significant advantages of smart contracts in improving transparency and fostering trust in e-commerce environments.

2.2. Live-streaming e-commerce

2.2.1. Origins and features of live-streaming e-commerce

In 2005, social commerce was first introduced on Yahoo.com, emphasizing the importance of consumer interaction during shopping (Jăşcanu et al., 2007). Social commerce is a sales method mediated through social media platforms (Shadkam and O'Hara, 2013). Kim and Park (2013) noted the scalability and innovation of social commerce in e-commerce, highlighting the importance of consumers in social networks. The live-streaming e-commerce ecosystem consists of several roles: live-streaming platforms (e.g., Taobao Live and Tiktok) provide technical support and traffic distribution, forming the core of the entire model; streamers and influencers drive sales by showcasing products and interacting with consumers in real-time; multi-channel network (MCN) agencies act as intermediaries, managing and training streamers and integrating resources; merchants and suppliers offer product support; and efficient logistics ensure a high-quality shopping experience.

2.2.2. Consumer behavior in live-streaming e-commerce

The rise of live-streaming e-commerce has transformed the traditional e-commerce landscape and introduced a new role for viewers as potential consumers. In traditional e-commerce, merchants rely on advertisements, celebrity endorsements, and electronic word-of-mouth to boost consumer purchase intentions (Manes and Tchetchik, 2018). However, the consumption environment in live-streaming e-commerce is more immersive and interactive, with streamers serving as the primary bridge for consumer interaction and decision-making.

Studies are increasingly focusing on the influence of streamer characteristics on consumer behavior in live streaming. Research showed that streamers who exhibit high levels of personal charm, credibility, interactivity, and authenticity are particularly effective in engaging viewers and fostering trust, thereby enhancing purchase intentions (Chen and Yang, 2023; Luo et al., 2024; Zheng et al., 2023). Furthermore, streamers who demonstrate expertise in product knowledge or maintain a transparent, authentic persona are more likely to form a connection with their audience, which significantly influences purchasing decisions (Ji et al., 2022; Yan et al., 2023; Zhang M. et al., 2022). The emergence of virtual technologies and artificial intelligence has also contributed to consumer engagement, with Gao et al. (2023) highlighting that virtual streamers can enhance trust and interaction, further shaping viewer perceptions and behaviors. Yuan et al. (2025) found that AI streamers can substitute human streamers in monetary engagement for utilitarian consumption, but their effect is short-term, while human streamers retain an advantage in non-monetary engagement. Moreover, in culturally specific contexts, Chen and Wu (2024) emphasize the role of social orientation and face consciousness in fostering emotional attachment and driving purchase intentions, particularly in esports live streaming where viewers are more likely to purchase virtual gifts. In the luxury brand domain, Li G. et al. (2023) found that the inclusion of internet celebrities and e-shop sellers in live streaming campaigns positively affects sales, with the streamer's level moderating these outcomes.

In addition to streamer characteristics, the live-streaming environment itself plays a crucial role in influencing consumer behavior. Elements such as bullet comments, real-time feedback, and group interactions create a highly interactive and social environment that strengthens consumers' emotional connection to the content. Bullet comments, in particular, provide immediate feedback on products, allowing viewers to see others' opinions and experiences, which can increase confidence in their own purchasing decisions (Li, B. et al., 2023). Research further indicates that the quantity, quality, and lexical diversity of bullet comments significantly impact consumer purchase intentions (Zeng et al., 2023; Zhang and Ruan, 2024). Positive interactions including enthusiastic comments or product endorsements from fellow viewers can generate a sense of urgency or social proof, thereby encouraging additional purchases (Bazi et al., 2023).

Overall, these immersive, socially interactive environments strongly influence consumer behavior. However, they also expose consumers to risks such as false advertising that exploit their emotional biases (Wu et al., 2023). For example, studies show that streamers' persuasive language and interactive behaviors create an emotionally engaging atmosphere, which can lead to impulsive purchasing decisions (Chen and Wu, 2024; Li L. et al., 2024; Zheng et al., 2023). Consequently, this emotional engagement makes consumers more susceptible to deceptive advertising, where misleading product claims capitalize on these biases.

2.2.3. Information asymmetry in live-streaming e-commerce

Information symmetry is a key challenge of live-streaming e-commerce, which can occur between suppliers and merchants (Wang, L. et al., 2024), and between merchants and consumers (Xu et al., 2022). In traditional e-commerce, merchants are typically responsible for advertising, product claims, and consumer trust-building. However, in live-streaming e-commerce, consumers must rely heavily on the streamer's recommendations and product descriptions to make purchasing decisions. This reliance creates significant gaps in information, leading to the potential for false advertising. For example, streamers may exaggerate product features or misrepresent product quality to boost sales, which can mislead consumers (Zhang et al., 2023).

To address these information asymmetry challenges, researchers have proposed various technical solutions. For instance, Li et al. (2021) developed a Live-streaming Fraud Detection model based on heterogeneous graph neural networks. By integrating heterogeneous information and employing label propagation algorithms, this model significantly improves fraud detection on live-streaming platforms. Furthermore, Liu et al. (2024) explored the adoption of blockchain technology in MCNs using a signaling model, examining its impact on retailer influencer selection and equilibrium conditions, offering a novel technological approach to mitigating information asymmetry. These studies highlight enhancing platform transparency and trust mechanisms through advanced technologies as a key strategy for addressing information asymmetry in live-streaming e-commerce.

Building on this, consumer purchase intentions are strongly shaped by emotional and cognitive biases, particularly within the immersive and social environments of these platforms. Although some scholars have begun exploring the issue of false advertising in live-streaming e-commerce, significant gaps remain in the literature. Notably, while blockchain technology is widely acknowledged as a promising tool with considerable potential, its application in live-streaming e-commerce remains underexplored. To address this gap, this study aims to examine the effectiveness of blockchain technology in mitigating the risks associated with false advertising in live-streaming e-commerce.

2.3. Theoretical foundation

2.3.1. Bounded rationality

Traditional economic and game theory often assume that individuals are fully rational decision-makers with unlimited cognitive abilities, complete information, and the capacity to optimize choices (Akerlof and Yellen, 1987; Conlisk, 1996). However, in real-world scenarios, decision-making is constrained by limited cognitive resources, incomplete information, and environmental complexities (Jones, 1999; Kahneman, 2003). In response to these limitations, Simon (1955) introduced the concept of bounded rationality, suggesting that individuals and organizations operate within cognitive and informational constraints, leading them to pursue satisficing rather than optimizing decisions. Bounded rationality implies that game participants cannot identify the optimal strategy from the outset, but refine their strategies through continuous learning and trial-and-error during interactions (Sugden, 1989).

In the live-streaming e-commerce environment, e-commerce platforms, merchants, and consumers interact within a dynamic ecosystem. Due to limited cognitive resources, incomplete information, and environmental uncertainties, these participants cannot immediately identify optimal strategies (Ji et al., 2023; Fan et al., 2024). Instead, they adjust their decisions based on past experiences, market feedback, and competitive pressures (Zhang et al., 2022; Zhang et al., 2025). For example, in the live-streaming e-commerce environment, a merchant initially promotes a product with exaggerated claims, leading to negative consumer feedback and penalties from the platform. Based on this feedback, the merchant shifts to more truthful advertising, showcasing real product results through customer reviews and demonstrations. This adjustment improves consumer trust, and positive reviews gradually increase, boosting sales. Consumers adjust their purchasing decisions based on these updated reviews, creating a dynamic feedback loop that enhances both trust and sales over time.

The bounded rationality of all participants leads them to rely on iterative learning and strategy adjustments, gradually converging toward equilibrium. In this context, platforms provide technical support and regulatory oversight, ensuring smooth transactions and protecting consumer rights (Zhang et al., 2023). Merchants adjust their promotional strategies, sometimes engaging in deceptive practices such as false advertising, which may conflict with platform regulations (Feng et al., 2021). Consumers, acting as both beneficiaries and market monitors, influence the system by providing feedback and defending their rights, such as through online complaints (Ren et al., 2024). These interactions form a dynamic game, where each participant continuously adapts to others' behavior.

To model the adaptive nature of bounded rationality, a method to simulate this process is required. Replicator dynamics, an evolutionary model based on group behavior, captures the diffusion of strategies and their adaptive adjustments over time (Roca et al., 2009). It effectively describes how individuals modify their strategies through repeated interactions and how these trends evolve (Adami et al., 2016). Given its ability to simulate adaptive learning in dynamic environments (Jiang et al., 2019; Lu and Zhang, 2022; Xue and Xiong, 2024), replicator dynamics is adopted in this study to model participants' strategic evolution in live-streaming e-commerce.

2.3.2. Evolutionary game theory

Traditional game theory assumes "perfect rationality" as its foundation but faces significant limitations in practical applications. These include difficulties in solving Nash equilibria, inadequate representation of dynamic behaviors in complex systems, and insufficient characterization of multiple equilibrium features. To address these shortcomings, evolutionary thinking and behavioral selection patterns in biology have been applied to game theory. Smith and Price (1973) studied the evolution and equilibrium of biological populations and introduced the evolutionary stable strategy (ESS) as a core concept, laying the foundation for evolutionary game theory (EGT). Building on this framework, Binmore (1988) explored the implications of EGT in economic and social interactions, highlighting its relevance beyond biological contexts. Friedman (1991) further examined the role of dynamic adaptation in strategic decision-making, emphasizing the learning processes that drive the evolution of strategies over time.

Unlike classical game theory, which assumes perfectly rational players with complete information, EGT accounts for bounded rationality and adaptive behavior, making it a vital tool for analyzing the dynamic behavior of complex systems (Dou et al., 2024). In recent years, this framework has been widely applied in e-commerce to investigate competitive and cooperative behaviors in dynamic environments. Considering the intricate interests of e-commerce stakeholders—such as suppliers, merchants, MCNs, influencers, and consumers—EGT effectively addresses multi-agent interactions and provides optimal equilibrium solutions. For example, Wu et al. (2020) demonstrated through EGT that consumer supervision and well-designed incentive mechanisms can reduce regulatory costs and mitigate issues like big data price discrimination. Similarly, Li C. et al. (2023) highlighted that government interventions such as enhanced regulation and clarified data ownership combined with consumer and media oversight can curb discriminatory pricing by platforms. Furthermore, Qiu et al. (2024) emphasized the significance of active regulation cost differentials, consumer compensation, and complaint costs in mitigating misconduct in live-streaming e-commerce.

While these studies provide valuable insights into regulatory challenges in live-streaming e-commerce, they focus on multi-agent interactions, overlooking how emerging technologies can reduce misconduct. Moreover, the inherent stochastic factors in live-streaming environments remain unexplored, highlighting the need to explore stochastic modeling approaches in regulatory mechanisms.

2.3.3. Itô stochastic differential equation theory

In practical game scenarios, stakeholders are often subject to significant internal and external disruptions (Jia et al., 2022). To address this, stochastic evolutionary game models, which incorporate random disturbance factors, offer distinct advantages in analyzing the effects of such disruptions on decision-making processes (Zhou et al., 2023). Here, the Itô stochastic differential equation is a powerful tool for modeling stochastic dynamic systems. It captures the dynamic evolution of stakeholders' behaviors under internal and external random disturbances.

Its general form is as follows:

$$dX_t = \mu(t, X_t)dt + \sigma(t, X_t)dW_t \tag{2.1}$$

Here, $\mu(t, X_t)$ represents the drift term, $\sigma(t, X_t)$ the diffusion term, and W_t the standard Brownian motion (a process with a mean of 0 and variance of t). Let $f(t, X_t)$ be a function with second-order continuous partial derivatives. Then, the derivation of the Itô formula is based on the Taylor expansion of the function. First, apply the Taylor expansion to $f(t, X_t)$:

$$f(t + dt, X_t + dX_t) \approx f(t, X_t) + \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial X_t}dX_t + \frac{1}{2}\frac{\partial^2 f}{\partial {X_t}^2}(dX_t)^2 \tag{2.2}$$

Given that X_t is an Itô process, substituting Equation (2.1) into Equation (2.2) and noting that the product of dt and dW_t is a higher-order infinitesimal term that can be ignored, we arrive at:

$$(dX_t)^2 = \sigma(t, X_t)^2 dt \tag{2.3}$$

Thus, the Itô formula can be expressed as:

$$df(t, X_t) = \frac{\partial f}{\partial t}dt + \frac{\partial f}{\partial X_t}(\mu(t, X_t)dt + \sigma(t, X_t)dW_t) + \frac{1}{2}\frac{\partial^2 f}{\partial {X_t}^2}\sigma(t, X_t)^2 dt \tag{2.4}$$

After rearrangement, we obtain:

$$df(t,X_t)=\left(\frac{\partial f}{\partial t}+\mu(t,X_t)\frac{\partial f}{\partial X_t}+\frac{1}{2}\frac{\partial^2 f}{\partial {X_t}^2}\sigma(t,X_t)^2\right)dt+\sigma(t,X_t)\frac{\partial f}{\partial X_t}dW_t \tag{2.5}$$

Previous studies applied stochastic differential equation theory to e-commerce supply chain and financial contexts. Guo and Chen (2024) examined the evolutionary dynamics of strategies in supply chain finance collaborations between e-commerce platforms and banks, focusing on the impact of random disturbances and contract parameters. Du et al. (2023) explored evolutionary equilibrium in supplier-influencer alliances, using white noise simulations to analyze the effects of parameter changes on stability. Zhu et al. (2023) developed a tripartite stochastic game model incorporating blockchain technology to address fraudulent behaviors in accounts receivable financing, investigating the effects of information sharing and regulatory intensity on strategic decisions. These studies provide a theoretical foundation for modeling complexity in the live-streaming e-commerce context.

Despite these advancements, the literature review reveals several gaps. Existing studies focus on static analyses of individual stakeholders or specific behaviors, and research on the dynamic interactions among multiple stakeholders remains limited. Furthermore, the impact of stochastic factors on strategic decision-making is underexplored, leaving the uncertainty inherent in the live-streaming e-commerce environment insufficiently addressed.

To address these gaps, this study innovatively constructs a tripartite evolutionary game model involving e-commerce platforms, merchants, and consumers in live-streaming supply chains, incorporating blockchain technology. By leveraging Itô stochastic differential equation theory and introducing stochastic disturbance factors, the model more accurately reflects the uncertainty characteristics of the live-streaming e-commerce environment. Through numerical simulations, this study examines the dynamic impacts of key parameters such as commission rates, benefits from false advertising, and the costs and benefits of blockchain technology. The findings provide theoretical insights and practical recommendations for promoting the sustainable development of the live-streaming e-commerce industry.

Fig. 1 illustrates the research framework of the stochastic evolutionary game approach for live-streaming e-commerce, providing a clear pathway for analyzing stakeholders' strategic evolution. Using the payoff matrix, this study examines the strategic choices of three key participants: e-commerce platforms, merchants, and consumers.

3. Model description and assumptions

3.1. Model description

Fig. 2 shows the live e-commerce supply chain tripartite game model, emphasizing the strategic interactions between the three key stakeholders: e-commerce platforms, merchants, and consumers.

E-commerce platforms provide merchants with technical support and regulatory oversight. Technical support includes advanced tools such as live streaming systems, payment gateways, traceability mechanisms (e.g., blockchain technology), and customer service platforms. These tools facilitate seamless transactions between merchants and consumers while enhancing operational transparency and service quality (Shen et al., 2021). Regulatory measures encompass actions to

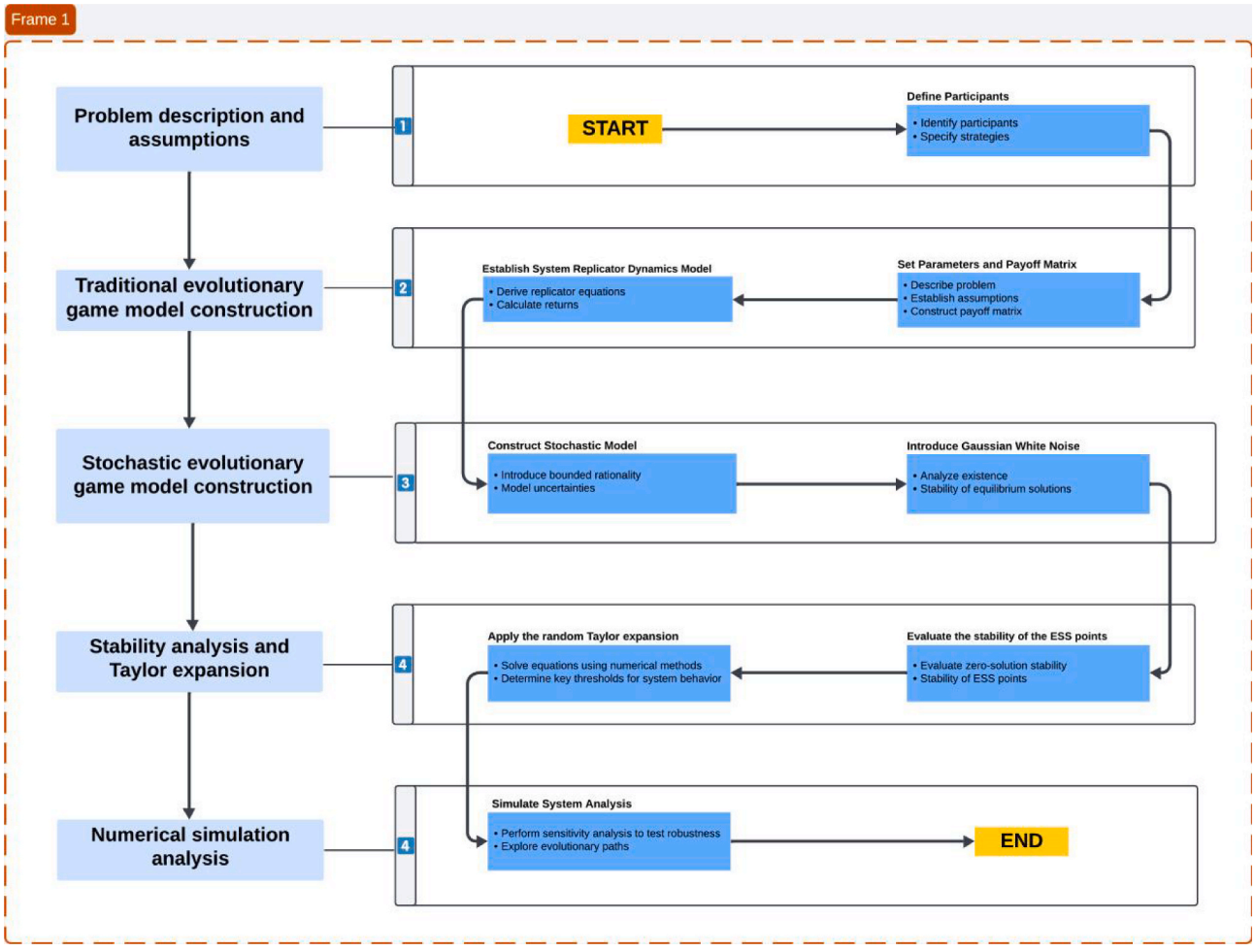


Fig. 1. Framework for tripartite stochastic evolutionary game analysis in live streaming E-commerce.

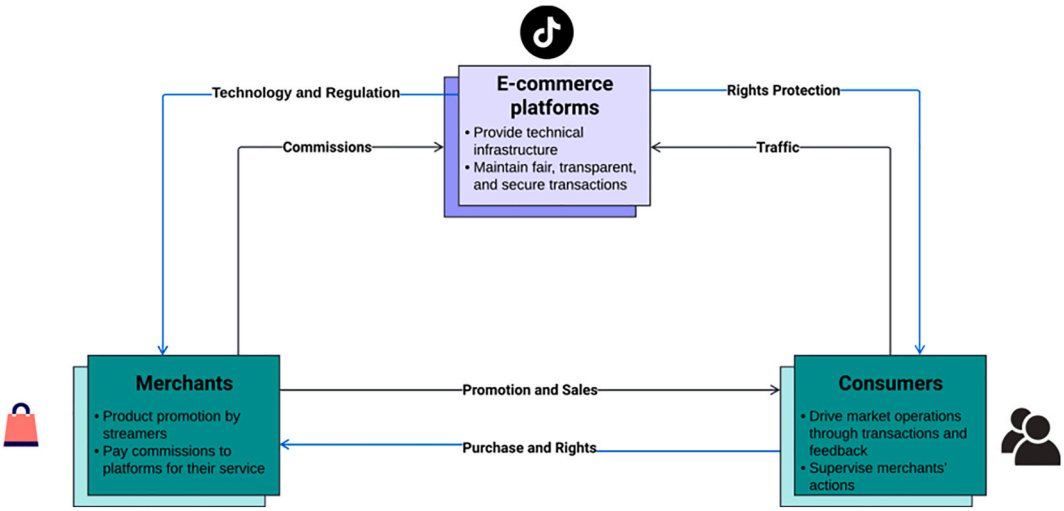


Fig. 2. Dynamic interactions among E-commerce platforms, merchants, and consumers.

ensure compliance and protect stakeholders' interests, including technical supervision, stricter penalties, and policy incentives like traffic allocation (Jia et al., 2022; Liu et al., 2024; Zhang, K. et al., 2024). The overarching goal is to regulate merchant behavior, safeguard consumer rights, and foster service recovery to create a secure, transparent, and efficient trading environment (Chen et al., 2023).

Merchants focus on promoting and selling goods while paying commissions to e-commerce platforms for access to resources and services (Wang Q. et al., 2024). By leveraging e-commerce platforms, particularly through highly interactive methods such as live streaming, streamers can utilize various promotional strategies to capture consumer attention and drive sales. The commission-based mechanism fosters a close, interdependent partnership between merchants and platforms. However, due to information asymmetry, there is a risk of collusion between merchants and suppliers, which may result in false advertising by streamers and pose significant challenges to consumer rights protection (Wang L. et al., 2024).

Consumers generate traffic for e-commerce platforms and profits for merchants by watching live streams and placing orders. As active participants and co-creators of value, they influence demand trends and provide valuable feedback to platforms and merchants (Zhang et al., 2023). Consumers protect their rights directly by offering feedback and monitoring product sales to uphold market fairness. For instance, they can post complaints on social media or leave real-time comments through live streaming barrage, negatively impacting merchants' sales (Zeng et al., 2023). Although consumers tend to prefer traceable products (Tran et al., 2024), their willingness to safeguard their rights may diminish under merchant inducements such as incentivized positive reviews (Xie et al., 2022).

In the above model, the cooperation and interactions between e-commerce platforms, merchants, and consumers reflect the dynamics and evolution of the e-commerce ecosystem. This involves a dynamic game where each party adjusts its strategies based on others' actions. E-commerce platforms provide technical support and regulatory measures to manage merchants' behavior and protect consumer rights. However, to maximize profits, merchants may face conflicts of interest with platforms, especially due to information asymmetry, which can lead to improper market behavior. Consumers, as both beneficiaries and market monitors, influence merchants through feedback and complaints. The relationship between these three parties creates a complex game, where each party continuously adapts to the others' actions. Accordingly, for this research, the following model assumptions are made:

(1) Participants: In this study, it is assumed that the three primary participants in live e-commerce are e-commerce platforms, merchants, and consumers.
(2) Behavioral assumptions: Due to limited information, bounded cognition, uncertain environments, changing preferences, and the instability of social values, all three participants exhibit bounded rationality (Di and Liu, 2016; Li T. et al., 2023). Consequently, they are unable to achieve optimal decisions immediately and must rely on constant learning and iterative processes to reach game equilibrium.
(3) Strategic choices: Each of the three participants has two strategic selections. E-commerce platforms choose between adopting and not adopting blockchain technology, with probabilities x and (1-x), respectively, where $0 \leq x \leq 1$. Merchants choose between truthful and false advertising, with probabilities y and (1-y), respectively, where $0 \leq y \leq 1$. Consumers choose between taking and not taking action to defend their rights, with probabilities z and (1- z), respectively, where $0 \leq z \leq 1$.

3.2. Parameter assumptions

Based on the assumptions and analytical framework outlined above, this study examines the significant impact of blockchain technology on the costs and benefits of interactions between e-commerce platforms, merchants, and consumers.

Regarding e-commerce platforms, according to Xing et al. (2022), the main source of platform revenue is the commission from merchants, set as a proportion of the merchant's revenue, denoted as a ($0 < a < 1$). In the blockchain technology context, the platform's development and operational costs are relatively high (Dehghani et al., 2022), denoted as C_B. In contrast, in the non-blockchain technology context, operational costs are lower, denoted as C_N. C_B includes: (1) technology development costs such as the blockchain technology network setup, smart contract development, and hardware and software support for data storage; and (2) operational and maintenance costs, including the operation of on-chain nodes, power consumption, and labor costs of the technical support team.

Regarding merchants, income is influenced by promotional strategies and e-commerce platforms' technology strategies. This is the expected revenue merchants typically generate under standard promotional strategies in a non-blockchain technology environment. This value is denoted as w in our model and serves as a reference point for evaluating the incremental revenue impact of blockchain technology adoption. This parameter is used to standardize comparisons and isolate

the effects of blockchain-related strategies on revenue growth. However, despite the potential risks, false advertising can boost sales (Yannopoulou et al., 2024), with this increase set as a proportion h ($0 < h$). Furthermore, increased supply chain transparency enhances consumer trust and improves brand image, which positively affects merchant income (Wang, H. et al., 2021; Duan and Zhu, 2024). Therefore, in the blockchain technology context, this study assumes that merchant income can increase by a factor of η ($0 < \eta < 1$) over the baseline income w. Blockchain technology enhances transparency and traceability, allowing platforms to better enforce penalties, denoted as F, on merchants for false advertising (Pun et al., 2021).

Regarding consumers, utility and the cost of defending rights change when blockchain technology is adopted. For products advertised truthfully, consumers gain utility U, whereas for falsely advertised products, the utility decreases to $U - I$, where I represents the utility loss due to false advertising. This loss can include monetary loss due to overpayment, harm to physical health from substandard products, and emotional distress caused by dissatisfaction (Piccolo et al., 2015). For non-blockchain technology, the consumer's cost of defending rights is C_t, whereas for blockchain technology, the traceability function reduces this cost to C_q ($C_q < C_t$). Furthermore, blockchain technology transparency increases consumer advocacy benefits (Boukis, 2019), raising it from G to G ($1+s$), where s represents the proportion of increased consumer awareness in defending their rights (Singh and Sharma, 2023).

To ensure the practical value of the study, we set the following boundary conditions for the model:

i. It applies to medium and large business-to-consumer live e-commerce platforms, assuming these platforms have sufficient resources and technological capability to adopt blockchain technology.
ii. We assume that brand merchants rely on platform traffic and only sell through online channels.
iii. Consumer rights protection costs and trust-enhancing effects are based on a specific cultural environment and industry (e.g., the food and luxury goods industries), where consumers prefer products with higher transparency.

For the purposes of this study, we simplified certain aspects. We ignore the diversity of deceptive advertising risks and do not consider the impact of competition between platforms. Additionally, our model assumes uniform regulatory enforcement across platforms; however, variations in regulatory stringency and market conditions may affect blockchain adoption feasibility across different regions (e.g., Asia vs. Europe). Furthermore, the estimates for η and s rely on a literature and case analysis, which may introduce subjective bias. Table 1 summarizes the symbols and their corresponding definitions used throughout the model, providing clarity and convenience for the subsequent analysis (see Table 2).

3.3. Model construction

For clarity and convenience, P_k, M_i, and C_j are used to denote the strategy choices of e-commerce platforms, merchants, and consumers, respectively. Specifically:

i. $k = 1$: E-commerce platforms opt not to adopt blockchain technology, while $k = 0$: Platforms adopt blockchain technology.
ii. $i = 1$: Merchants select a truthful advertising strategy, while $i = 0$: Merchants choose a false advertising strategy.
iii. $j = 1$: Consumers decide to defend their rights, while $j = 0$: Consumers opt to not defend their rights.

Scenario (k, i, j): (1, 1, 1).

When e-commerce platforms do not adopt blockchain technology, merchants choose truthful advertising, and consumers choose to defend their rights, the incomes of the three parties are:

$$\begin{aligned} \pi_{P11} &= aw - C_N \\ \pi_{M11} &= w(1-a) \\ \pi_{C11} &= U - C_t \end{aligned} \tag{3.1}$$

Scenario (k, i, j): (1, 1, 0).

When e-commerce platforms do not adopt blockchain technology, merchants choose truthful advertising, and consumers choose not to defend their rights, the incomes of the three parties are:

$$\begin{aligned} \pi_{P21} &= aw - C_N \\ \pi_{M21} &= w(1-a) \\ \pi_{C21} &= U \end{aligned} \tag{3.2}$$

Scenario (k, i, j): (1, 0, 1).

When e-commerce platforms do not adopt blockchain technology, merchants choose false advertising, and consumers choose to defend their rights, the incomes of the three parties are:

$$\begin{aligned} \pi_{P31} &= aw(1+h) - C_N \\ \pi_{M31} &= w(1-a)(1+h) - G \\ \pi_{C31} &= U - I - C_t + G \end{aligned} \tag{3.3}$$

Scenario (k, i, j): (1, 0, 0).

When e-commerce platforms do not adopt blockchain technology, merchants choose false advertising, and consumers choose not to defend their rights, the incomes of the three parties are:

$$\begin{aligned} \pi_{P41} &= aw(1+h) - C_N \\ \pi_{M41} &= w(1-a)(1+h) \\ \pi_{C41} &= U - I \end{aligned} \tag{3.4}$$

Scenario (k, i, j): (0, 1, 1).

When e-commerce platforms adopt blockchain technology, merchants choose truthful advertising, and consumers choose to defend their rights, the incomes of the three parties are:

Table 1
Parameters and their descriptions.

Parameters	Descriptions
a	Commission rate on brand revenue
C_B	Cost for e-commerce platform in the blockchain technology context
C_N	Cost for e-commerce platform in the non-blockchain technology context
w	Benchmark revenue for merchants under baseline conditions
F	Penalty for false advertising by brand
h	Sales growth factor attributable to false advertising
η	Proportionate increase in sales revenue in the blockchain technology context
U	Consumer utility derived from purchasing real advertised goods
I	Perceived consumer utility loss from falsely advertised goods
C_t	Cost of consumers defending their rights in the non-blockchain technology context
C_q	Cost of consumers defending their rights in the blockchain technology context
G	Consumer gain from reporting false advertising
s	Proportionate increase in consumer awareness of defending their rights due to blockchain technology
π_P	Payoff function representing the income of the e-commerce platform
π_M	Payoff function representing the income of the brand
π_C	Payoff function representing the income of the consumer

Table 2
Description of variables.

Variable	Descriptions
X	Probability that e-commerce platforms choose not to adopt blockchain technology
Y	Probability that merchants choose to engage in false advertising
Z	Probability that consumers choose to defend their rights

$$\begin{aligned}\pi_{P12} &= aw(1+\eta) - C_B \\ \pi_{M12} &= w(1-a)(1+\eta) \\ \pi_{C12} &= U - C_q\end{aligned} \tag{3.5}$$

Scenario (*k*, *i*, *j*): (0, 1, 0).

When e-commerce platforms adopt blockchain technology, merchants choose truthful advertising, and consumers choose not to defend their rights, the incomes of the three parties are:

$$\begin{aligned}\pi_{P22} &= aw(1+\eta) - C_B \\ \pi_{M22} &= w(1-a)(1+\eta) \\ \pi_{C22} &= U\end{aligned} \tag{3.6}$$

Scenario (*k*, *i*, *j*): (0, 0, 1).

When e-commerce platforms adopt blockchain technology, merchants choose false advertising, and consumers choose to defend their rights, the incomes of the three parties are:

$$\begin{aligned}\pi_{P32} &= aw(1+h)(1+\eta) - C_B + F \\ \pi_{M32} &= w(1-a)(1+h)(1+\eta) - G(1+s) - F \\ \pi_{C32} &= U - I - C_q + G(1+s)\end{aligned} \tag{3.7}$$

Scenario (*k*, *i*, *j*): (0, 0, 0).

When e-commerce platforms adopt blockchain technology, merchants choose false advertising, and consumers choose not to defend their rights, the incomes of the three parties are:

$$\begin{aligned}\pi_{P42} &= aw(1+h)(1+\eta) - C_B + F \\ \pi_{M42} &= w(1-a)(1+h)(1+\eta) - F \\ \pi_{C42} &= U - I\end{aligned} \tag{3.8}$$

Based on the assumed parameters and variable settings, Table 3 presents the payoff benefits for e-commerce platforms, merchants, and consumers under different strategy combinations.

According to the game returns under each strategy in Table 3, the expected returns under different strategies are calculated.

(1) U_{P1} and U_{P2} represent the expected returns when e-commerce platforms choose not to adopt or adopt blockchain technology, respectively. U_P denotes the average expected returns of e-commerce platforms, with their calculation results expressed as follows:

$$\begin{aligned}U_{P1} = {} & yz(aw - C_N) + y(1-z)(aw - C_N) + (1-y)z(aw(1+h) - C_N) \\ & + (1-y)(1-z)(aw(1+h) - C_N)\end{aligned} \tag{3.9}$$

$$\begin{aligned}U_{P2} = {} & yz(aw - C_B)(1+\eta) + y(1-z)(aw - C_B)(1+\eta) + (1-y)z(aw(1+\eta)(1+h) - C_B + F) \\ & + (1-y)(1-z)(aw(1+\eta)(1+h) - C_B + F)\end{aligned} \tag{3.10}$$

$$U_P = xU_{P1} + (1-x)U_{P2} \tag{3.11}$$

(2) U_{M1} and U_{M2} represent the expected returns when merchants choose false advertising or not, respectively. U_M denotes the average expected returns of merchants, with their calculation results expressed as follows:

$$\begin{aligned}U_{M1} = {} & xzw(1-a) + x(1-z)w(1-a) + (1-x)zw(1-a)(1+\eta) \\ & + (1-x)(1-z)w(1-a)(1+\eta)\end{aligned} \tag{3.12}$$

$$\begin{aligned}U_{M2} = {} & xz(w(1-a)(1+h) - G) + x(1-z)w(1-a)(1+h) \\ & + (1-x)z(w(1-a)(1+h)(1+\eta) - G(1+s) - F) + (1-x)(1-z)(w(1-a)(1+h)(1+\eta) - F)\end{aligned} \tag{3.13}$$

$$U_M = yU_{M1} + (1-y)U_{M2} \tag{3.14}$$

(3) U_{C1} and U_{C2} represent the expected returns when consumers choose to defend their rights or not, respectively. U_C denotes the average expected returns of consumers, with their calculation results expressed as follows:

$$\begin{aligned}U_{C1} = {} & xy(U - C_t) + x(1-y)(U - I + G - C_t) + (1-x)y(U - C_q) \\ & + (1-x)(1-y)(U - I - C_q + G(1+s))\end{aligned} \tag{3.15}$$

$$U_{C2} = xyU + x(1-y)(U-I) + (1-x)yU + (1-x)(1-y)(U-I) \tag{3.16}$$

$$U_C = zU_{C1} + (1-z)U_{C2} \tag{3.17}$$

Further calculations show that the replication dynamic equations for e-commerce platforms, merchants, and consumers are $F_1(x)$, $F_2(y)$, and $F_3(z)$:

$$F_1(x) = \frac{dx}{dt} = x(U_{P1} - U_P) = x(1-x)(U_{P1} - U_{P2}) \tag{3.18}$$

$$F_2(y) = \frac{dy}{dt} = y(U_{M1} - U_M) = y(1-y)(U_{M1} - U_{M2}) \tag{3.19}$$

$$F_3(z) = \frac{dz}{dt} = z(U_{C1} - U_C) = z(1-z)(U_{C1} - U_{C2}) \tag{3.20}$$

From $0 \le x, y, z \le 1$, we know that 1-x, 1-y, 1-z are non-negative numbers, which will not affect the stability of the strategy. Therefore, following Sun et al. (2016), the replication dynamics equations are adjusted accordingly. The adjusted tripartite replication dynamic equations are as follows:

$$\begin{aligned}F_1(x) &= \frac{dx}{dt} = x(U_{P1} - U_{P2}) \\ &= x[(C_B\eta - C_N\eta)yz + y(F + a\eta w(1+h)) + C_B - C_N - F - a\eta w(1+h)]\end{aligned} \tag{3.21}$$

Table 3
Payoff matrix.

Scenario	Payoff matrix	Scenario	Payoff matrix
(1, 1, 1)	$(\pi_{P11}, \pi_{M11}, \pi_{C11})$	(0, 1, 1)	$(\pi_{P12}, \pi_{M12}, \pi_{C12})$
(1, 1, 0)	$(\pi_{P21}, \pi_{M21}, \pi_{C21})$	(0, 1, 0)	$(\pi_{P22}, \pi_{M22}, \pi_{C22})$
(1, 0, 1)	$(\pi_{P31}, \pi_{M31}, \pi_{C31})$	(0, 0, 1)	$(\pi_{P32}, \pi_{M32}, \pi_{C32})$
(1, 0, 0)	$(\pi_{P41}, \pi_{M41}, \pi_{C41})$	(0, 0, 0)	$(\pi_{P42}, \pi_{M42}, \pi_{C42})$

$$F_2(y)=\frac{dy}{dt}=y(U_{M1}-U_{M2}) \\ =y[(G(2x-1)+(2w(a-1)(1+\eta)(1+h)-2F)(1-x))z+F(1-x)+w(1-a)(\eta hx-h-\eta h)] \tag{3.22}$$

$$F_3(z)=\frac{dz}{dt}=z(U_{C1}-U_{C2}) \\ =z\left[G(1+s)(1-y)+\left(G_q-C_t-Gs(1-y)\right)x\right] \tag{3.23}$$

3.4. Replication dynamic equation

When a strategy generates higher-than-average returns, it stabilizes in the evolutionary process over time, as explained by replication dynamics theory (Kang et al., 2023; Wei and She, 2023). For example, if blockchain technology yields higher returns for an e-commerce platform, the platform is more likely to adopt it, increasing the prevalence of this behavior. Similarly, merchants and consumers will dynamically adjust their strategies based on their respective payoffs. In the following sections, we solve replication dynamic equations for the three-party strategies. Replicator dynamic equations are developed for three heterogeneous groups—e-commerce platforms, merchants, and consumers—in the context of governing false advertising events facilitated by blockchain technology.

4. Construction of stochastic evolutionary game model

4.1. Modeling theorem

In the practical live-streaming e-commerce environment, factors such as market demand (Wang et al., 2022), supply chain risks (Zhang X. et al., 2024), hosts' efforts and performance (Zhang et al., 2022), and platform traffic distribution (Zhang Y. et al., 2024) are subject to significant uncertainty. Each party does not always have accurate access to other parties' information. Merchants may not be able to fully understand consumer preferences, and consumers may not be able to fully understand the pricing strategies of e-commerce platforms or merchants. This incomplete information leads to uncertainty about other participants' future market movements and behavior. Therefore, the potential impact of stochastic perturbations on the three-party game must be considered. Introducing Itô stochastic differential equations into evolutionary game models involving three parties can more accurately capture the complexity of real market dynamics.

Gaussian white noise can be introduced to represent random disturbances in the system. This is a random signal with variations that follow a normal distribution, having a mean of zero and equal intensity across all frequencies (Yang and Mo, 2021). This feature makes it suitable for modeling unpredictable and irregular factors that influence the strategic decisions of multi-agents in the governance of false advertising.

Gaussian white noise is introduced into Equations (3.7), (3.8), and (3.9), and the model is modified by adding the stochastic perturbation term as follows:

$$dx=[(C_B\eta-C_N\eta)yz+y(F+a\eta w(1+h))+C_B-C_N-F-a\eta w(1+h)]x(t)dt \\ +\sigma x(t)dw(t) \tag{4.1}$$

$$dy=[(G(2x-1)+(2w(a-1)(1+\eta)(1+h)-2F)(1-x))z+F(1-x) \\ +w(1-a)(\eta hx-h-\eta h)]y(t)dt+\sigma y(t)w(t) \tag{4.2}$$

$$dz=\left[G(1+s)(1-y)+\left(G_q-C_t-Gs(1-y)\right)x\right]z(t)dt+\sigma z(t)dw(t) \tag{4.3}$$

Equations (4.1), (4.2), and (4.3) are one-dimensional Itô stochastic differential equations, which represent the dynamic equations of the evolutionary replication of e-commerce platforms, merchants, and consumers exposed to stochastic perturbations, respectively. Here, $w(t)$ is a one-dimensional standard Brownian motion, an irregular stochastic rise-and-fall phenomenon that reflects how stochastic disturbances in complexity factors influence the game decisions of multi-agents involved in addressing false advertising (Traulsen and Nowak, 2006). $dw(t)$ denotes Gaussian white noise, when the given time $t>0$ and step size $l>0$, $w(t)$ and its incremental $\Delta w(t)=w(t+l)-w(t)$ obeys the normal distribution $N(0,\sqrt{l})$. $\sigma=\sqrt{x(1-x)}$ represents the strength of the random disturbance, denoted as $\sigma\le\frac{1}{2}$. When $\sigma=\frac{1}{2}$, the random disturbance reaches its maximum, rendering the system most unstable (Shan et al., 2023).

4.2. Existence and stability analysis of the equilibrium solution of the stochastic game model

For one-dimensional stochastic differential equations based on Itô calculus (Wagner, 1982), assuming the initial time of the game is $t=0$, with $x(0)=0$, $y(0)=0$, $z(0)=0$, the following equations are derived:

$$[(C_B\eta-C_N\eta)yz+y(F+a\eta w(1+h))+C_B-C_N-F-a\eta w(1+h)]\cdot 0 \\ +\sigma x(t)dw(t)=0 \tag{4.4}$$

$$[(G(2x-1)+(2w(a-1)(1+\eta)(1+h)-2F)(1-x))z+F(1-x)+w(1-a)(\eta hx-h-\eta h)]\cdot 0 \\ +\sigma y(t)dw(t)=0 \tag{4.5}$$

$$\left[G(1+s)(1-y)+\left(G_q-C_t-Gs(1-y)\right)x\right]\cdot 0+\sigma z(t)dw(t)=0 \tag{4.6}$$

The above three equations with zero values clarify that at least a zero solution to the equation exists as follows:

$$dw(t)|t=0=w'(t)dt|t=0 \tag{4.7}$$

The existence of at least a zero solution to the equation indicates that the system will remain in that state in the absence of interference from external white noise. Therefore, the zero solution is an equilibrium solution of the equation.

The stability of a three-way evolutionary game equation can be assessed based on the stability discrimination theorem for stochastic differential equations (Adami et al., 2016; Moghaddam and Akbari,

2018; Zhao et al., 2024) as follows:

For a given differential equation:

$$dx(t) = f(t, x(t))dt + g(t, x(t))dw(t), x(t_0) = x_0 \tag{4.8}$$

Assume there exists function $V(t, x)$ and two positive constants c_1, c_2, which enable:

$$c_1|x|_p \le V(t, x) \le c_2|x|_p, t \ge 0. \tag{4.9}$$

Case 1.

If a positive constant γ exists, which enables $(t, x) \le -\gamma V(t, x), t \ge 0$, then the zero solution of Equation (4.8) is exponentially stable in order moments and holds that:

$$E|x(t, x_0)|_p \le \frac{c_2}{c_1}|x_0|_p e^{\gamma t}, t \ge 0. \tag{4.10}$$

Case 2.

If a positive constant γ exists, which enables $(t, x) \ge -\gamma V(t, x), t \le 0$, then the zero solution of Equation (4.8) is exponentially stable in order moments and holds that:

$$E|x(t, x_0)|_p > \frac{c_2}{c_1}|x_0|_p e^{\gamma t}, t \le 0. \tag{4.11}$$

To replicate the dynamic equations in (4.1), (4.2), and (4.3), the values are as follows:

$$V_t(t, x) = x, V_t(t, y) = y, V_t(t, z) = z; x, y, z \in [0, 1], c_1 = c_2 = 1, p = 1, \gamma = 1$$

We obtain:

$$LV(t, x) = x[(C_B\eta - C_N\eta)yz + y(F + a\eta w(1 + h)) + C_B - C_N - F - a\eta w(1 + h)] \tag{4.12}$$

$$LV(t, y) = y[(G(2x - 1) + (2w(a - 1)(1 + \eta)(1 + h) - 2F)(1 - x))z + F(1 - x) + w(1 - a)(\eta hx - h - \eta h)] \tag{4.13}$$

$$LV(t, z) = z\left[G(1 + s)(1 - y) + \left(G_q - C_t - Gs(1 - y)\right)x\right] \tag{4.14}$$

If the zero solution moments of the three equations $LV(t, x)$, $LV(t, y)$, and $LV(t, z)$ are exponentially stable, then collating Equations (4.1), (4.2)and (4.3) yields:

$$x[(C_B\eta - C_N\eta)yz + y(F + a\eta w(1 + h)) + C_B - C_N - F - a\eta w(1 + h)] \le -x \tag{4.15}$$

$$y[(G(2x - 1) + (2w(a - 1)(1 + \eta)(1 + h) - 2F)(1 - x))z + F(1 - x) + w(1 - a)(\eta hx - h - \eta h)] \le -y \tag{4.16}$$

$$z\left[G(1 + s)(1 - y) + \left(G_q - C_t - Gs(1 - y)\right)x\right] \le -z \tag{4.17}$$

Approximate reduction of the above three inequalities:

(1) For inequality (4.15), when $x \in [0, 1]$,

$$y \le \frac{F - C_B + C_N - 1 + a\eta w(1 + h)}{(C_B\eta - C_N\eta)z + F + a\eta w(1 + h)} \tag{4.18}$$

Denote inequality (4.18) as A.

(2) For inequality (4.16),

Case 1.

When $y \in [0, 1]$, with $G(2x - 1) + (2w(a - 1)(1 + \eta)(1 + h) - 2F)(1 - x) > 0$, we obtain:

$$z \le \frac{F(x - 1) + w(a - 1)(\eta hx - h - \eta h) - 1}{G(2x - 1) + (2w(a - 1)(1 + \eta)(1 + h) - 2F)(1 - x)} \tag{4.19}$$

Denote inequality (4.19) as B_1.

Case 2.

When $y \in [0, 1]$, with $G(2x - 1) + (2w(a - 1)(1 + \eta)(1 + h) -$

Table 4
Stability evaluation of equilibrium points.

Equilibrium Points	Eigenvalues	Stability Evaluation
E_1 (1, 1, 1)	$\lambda_1 = -G - hw(a - 1)$ $\lambda_2 = C_t - C_q$ $\lambda_3 = -(C_B - C_N)(F\eta + 1)$	Due to $C_t - C_q > 0$, this SEP is unstable
E_2 (1, 1, 0)	$\lambda_1 = G + hw(a - 1)$ $\lambda_2 = C_t - C_q - G$ $\lambda_3 = C_N - C_B + F + a\eta w(1 + h)$	Due to $-hw(a - 1) > 0$, this SEP is unstable
E_3 (1, 0, 1)	$\lambda_1 = C_N - C_B$ $\lambda_2 = C_q - C_t$ $\lambda_3 = -hw(a - 1)$	Stability requires certain conditions
E_4 (1, 0, 0)	$\lambda_1 = C_q - C_t + G$ $\lambda_2 = hw(a - 1)$ $\lambda_3 = C_N - C_B + F + a\eta w(1 + h)$	Stability requires certain conditions
E_5 (0, 1, 1)	$\lambda_1 = 0$ $\lambda_2 = (C_B - C_N)(F\eta + 1)$ $\lambda_3 = F + G - 2w(a - 1)(\eta + 1)(h + 1) - hw(a - 1)(\eta + 1)$	Due to $(C_B - C_N)(F\eta + 1) > 0$, this SEP is unstable
E_6 (0, 1, 0)	$\lambda_1 = 0$ $\lambda_2 = C_B - C_N$ $\lambda_3 = -F - w(a - 1)(h + \eta h)$	Due to $C_B - C_N > 0$, this SEP is unstable
E_7 (0, 0, 1)	$\lambda_1 = -G(s + 1)$ $\lambda_2 = 2w(a - 1)(\eta + 1)(h + 1) - G - F + hw(a - 1)(\eta + 1)$ $\lambda_3 = C_B - C_N - F - a\eta w - a\eta hw$	Stability requires certain conditions
E_8 (0, 0, 0)	$\lambda_1 = F + w(a - 1)(h + \eta h)$ $\lambda_2 = G(s + 1)$ $\lambda_3 = C_B - C_N - F - a\eta w - ah\eta w$	Due to $G(s + 1) > 0$, this SEP is unstable

$2F)(1-x)<0$, we obtain:

$$z \geq \frac{F(x-1)+w(a-1)(\eta hx-h-\eta h)-1}{G(2x-1)+(2w(a-1)(1+\eta)(1+h)-2F)(1-x)} \tag{4.20}$$

Denote inequality (4.20) as B_2.

(3) For inequality (4.17),

When $z \in [0,1]$,
Case 1.
When $z \in [0,1]$, with $G_q - C_t - Gs(1-y) > 0$, we obtain:

$$x \leq \frac{-1-G(1+s)(1-y)}{G_q - C_t - Gs(1-y)} \tag{4.21}$$

Denote this inequality (4.21) as C_1.
Case 2.
When $z \in [0,1]$, with $G_q - C_t - Gs(1-y) < 0$, we obtain:

$$x \geq \frac{-1-G(1+s)(1-y)}{G_q - C_t - Gs(1-y)} \tag{4.22}$$

Denote this inequality (4.22) as C_2.

The analysis of the zero-solution p-th moment exponential stability conditions demonstrates that under parameter conditions $A \cap (B_1 \cup B_2) \cap (C_1 \cup C_2)$, the strategy set (0, 0, 0) is stable. This indicates that the e-commerce platform adopts blockchain technology, merchants engage in false advertising, and consumers choose not to defend their rights, forming a behavior pattern that remains stable under stochastic disturbances. Even with blockchain technology as a regulatory mechanism, if the long-term benefits of false advertising for merchants outweigh the penalties and reputational constraints imposed by consumer complaints, the issue may persist.

4.3. Stochastic Taylor expansion of evolutionary game model

To better explain the solution, we apply the stochastic Taylor expansion to approximate the stochastic differential equation. The Taylor expansion is a formula that uses information about a function at a given point to describe its behavior in the neighborhood of that point. Given the complexity of the function, it is not feasible to directly seek the values at specific points. Therefore, we use the Taylor expansion to approximate the values for e-commerce platforms, merchants, and consumers (Liu et al., 2021).

Based on the previous assumptions of Equation (4.8), we further assume $h = \frac{T-t_0}{N}, t_n = t_0 + nh$ to obtain the Taylor expansion as follows:

$$x(t_{n+1}) = x(t_n) + I_0 f(x(t_n)) + I_1 g(x(t_n)) + I_{11} L^1 g(x(t_n)) + I_{00} L^0 f(x(t_n)) + R \tag{4.23}$$

Where $L^0 = f(x)\frac{\partial}{\partial x} + \frac{1}{2}g^2(x)\frac{\partial^2}{\partial x^2}, L^1 = g(x)\frac{\partial}{\partial x}, I_0 = h, I_1 = \Delta w_n, I_{00} = \frac{1}{2}h^2$, $I_{11} = \frac{1}{2}[(\Delta w_n)^2 - h]$ and R represents the remainder of the Taylor expansion. Equation (4.23) can then be rearranged and expressed as follows:

$$\begin{aligned} x(t_{n+1}) = x(t_n) + hf(x(t_n)) + \Delta w_n g(x(t_n)) + \frac{1}{2}\left[(\Delta w_n)^2 - h\right] g(x(t_n)) g'(x(t_n)) \\ + \frac{1}{2}h^2\left[f(x(t_n))f'(x(t_n)) + \frac{1}{2}g^2(x(t_n))f''(x(t_n))\right] + R \end{aligned} \tag{4.24}$$

We apply the Milstein numerical model to address this issue. Accordingly, the Taylor expansion for e-commerce platforms, merchants, and consumers under central government regulation is expressed as follows:

$$\begin{aligned} x(t_{n+1}) = x(t_n) + h[(C_B\eta - C_N\eta)yz + y(F + a\eta w(1+h)) + C_B - C_N - F - a\eta w(1+h)]x(t_n) + \Delta w_n \sigma x(t_n) + \frac{1}{2}\left[(\Delta w_n)^2 - h\right]\sigma^2 x(t_n) \\ + \frac{1}{2}h^2[(C_B\eta - C_N\eta)yz + y(F + a\eta w(1+h)) + C_B - C_N - F - a\eta w(1+h)]^2 x(t_n) + R_1 \end{aligned} \tag{4.25}$$

$$\begin{aligned} y(t_{n+1}) = y(t_n) + h[G(2x-1) + (2w(a-1)(1+\eta)(1+h) - 2F)(1-x))z + F(1-x) + w(1-a)(\eta hx - h - \eta h)]y(t_n) + \Delta w_n \sigma y(t_n) \\ + \frac{1}{2}\left[(\Delta w_n)^2 - h\right]\sigma^2 y(t_n) + \frac{1}{2}h^2[(G(2x-1) + (2w(a-1)(1+\eta)(1+h) - 2F)(1-x))z + F(1-x) + w(1-a)(\eta hx - h - \eta h)]^2 y(t_n) + R_2 \end{aligned} \tag{4.26}$$

$$\begin{aligned} z(t_{n+1}) = z(t_n) + h\left[G(1+s)(1-y) + C_q - C_t - Gs(1-y)\right)x]z(t_n) + \Delta w_n \sigma z(t_n) \\ + \frac{1}{2}\left[(\Delta w_n)^2 - h\right]\sigma^2 z(t_n) + \frac{1}{2}h^2\left[G(1+s)(1-y) + C_q - C_t - Gs(1-y)\right)x\right]^2 z(t_n) + R_3 \end{aligned} \tag{4.27}$$

With this method, we derive numerical approximations for e-commerce platforms, merchants, and consumers, resolving the challenge of lacking explicit solutions and numerical methods for Itô's stochastic differential equations.

4.4. Extended stability analyses of the stochastic evolutionary game model

In section 4.3, we analyzed the zero-solution p-th moment exponential stability, identifying conditions under which certain strategy combinations remain stable under stochastic disturbances. To extend this analysis, we introduce ESS to explore the stability and rationality of other equilibrium points using the Jacobian matrix.

Let $F_1 = 0$, $F_2 = 0$, and $F_3 = 0$, where nine points of equilibrium exist by solving the group of replicator dynamic equations, E_1 (1, 1, 1), E_2 (1, 1, 0), E_3 (1, 0, 1), E_4 (1, 0, 0), E_5 (0, 1, 1), E_6 (0, 1, 0), E_7 (0, 0, 1), E_8 (0, 0, 0), and E_9 (x_0, y_0, z_0). According to Li Y. et al. (2024) and Ritzberger and Weibull (1995), E_9 (x_0, y_0, z_0) represents a mixed-strategy equilibrium for the three-dimensional dynamic system, which cannot qualify as an ESS.

For the eight remaining boundary points, an equilibrium point qualifies as an ESS when all eigenvalues of its Jacobian matrix are negative. A saddle point arises if one or two eigenvalues are positive, while a point becomes unstable if all eigenvalues are positive. The first step in verifying the stability of these equilibrium points is to compute the Jacobian matrix for the tripartite evolution system.

Based on the previous calculation of Equation (3.18)–(3.20), we obtain the Jacobian matrix as follows:

$$J=\begin{bmatrix}\frac{\partial F_1}{\partial x} & \frac{\partial F_1}{\partial y} & \frac{\partial F_1}{\partial z}\\ \frac{\partial F_2}{\partial x} & \frac{\partial F_2}{\partial y} & \frac{\partial F_2}{\partial z}\\ \frac{\partial F_3}{\partial x} & \frac{\partial F_3}{\partial y} & \frac{\partial F_3}{\partial z}\end{bmatrix} \tag{4.28}$$

In the above matrix:

$$\frac{\partial F_1}{\partial x}=(1-2x)\left((C_B-C_N)2F\eta y^2z^2+F-a\eta w(1+h)(1-y)\right)$$
$$\frac{\partial F_1}{\partial y}=-x(x-1)\left(2F\eta yz^2(C_B-C_N)+F+a\eta w(h+1)\right)$$
$$\frac{\partial F_1}{\partial z}=-2F\eta xy^2z(C_B-C_N)(x-1)$$
$$\frac{\partial F_2}{\partial x}=y(y-1)(F-z(2F+2G-2w(a-1)(\eta+1)(h+1))+\eta hw(a-1))$$
$$\frac{\partial F_2}{\partial y}=(1-2y)(G(2x-1)+(2w(a-1)(1+\eta)(1+h)-2F)(1-x)z+F(1-x)+w(1-a)(\eta hx-h-\eta h))$$
$$\frac{\partial F_2}{\partial z}=-y((2F-2w(a-1)(\eta+1)(h+1))(x-1)+G(2x-1))(y-1)$$
$$\frac{\partial F_3}{\partial x}=-z(z-1)\left(C_q-C_t+Gs(y-1)\right)$$
$$\frac{\partial F_3}{\partial y}=Gz(z-1)(s-sx+1)$$
$$\frac{\partial F_3}{\partial z}=(1-2z)\left(G(1+s)(1-y)+\left(C_q-C_t-Gs(y-1)\right)x\right)$$

The Lyapunov first method can be used to evaluate the stability of each equilibrium point (Zhao et al., 2024). If any eigenvalue of the Jacobian matrix has a positive real part, the strategy equilibrium point (SEP) becomes unstable. Conversely, when all eigenvalues possess negative real parts, the SEP achieves asymptotic stability, thereby establishing an ESS. Table 4 presents the eigenvalues under different SEP conditions and provides a stability evaluation.

According to Table 4, three SEPs demonstrate stability under certain conditions: E_3 (1, 0, 1), E_4 (1, 0, 0), and E_7 (0, 0, 1). Next, we analyze the specific conditions that stabilize these SEPs to gain deeper insights into the stability of interactions among the three parties: e-commerce platforms, merchants, and consumers.

Case 1.

Case 1: Stability of E_3 (1, 0, 1) under specific conditions:

$$G<hw(1-a), C_t-C_q<G, F+a\eta w(1+h)<C_B-C_N \tag{4.29}$$

In equilibrium E_3 (1, 0, 1), e-commerce platforms opt not to adopt blockchain technology, merchants engage in false advertising, and consumers choose to defend their rights. This dynamic emerges because the costs of blockchain technology adoption C_B outweigh its benefits for platforms, while penalties F and increased sales from blockchain technology $a\eta w$ (1+h) fail to deter merchants from false advertising. When the impact of blockchain technology on merchants cannot offset the profits gained from false advertising, its role in combating misinformation diminishes significantly.

Case 2: Stability of E_4 (1, 0, 0) under specific conditions:

$$G<C_t-C_q, F+a\eta w(1+h)<C_B-C_N \tag{4.30}$$

In this scenario, equilibrium E_4 (1, 0, 0) becomes stable when consumers' gains from reporting false advertising G are less than the difference in the costs associated with defending their rights between non-blockchain technology and blockchain technology systems C_t-C_q. This indicates that the incentive for consumers to actively defend their rights diminishes when their cost savings from blockchain technology adoption are insufficient to offset the gains from reporting.

Furthermore, condition $F+a\eta w(1+h)<C_B-C_N$ highlights that the additional costs of e-commerce platforms adopting blockchain technology outweighs the benefits of enforcing penalties on merchants and leveraging increased consumer demand under blockchain technology $a\eta w(1+h)$. This cost-benefit imbalance discourages the platform from adopting this technology, leading merchants to maintain false advertising strategies without regulatory constraints.

Here, the interactions among the three parties reflect a lack of sufficient incentives for both platforms and consumers to support the defense of rights based on blockchain technology, ultimately stabilizing E_4 (1, 0, 0).

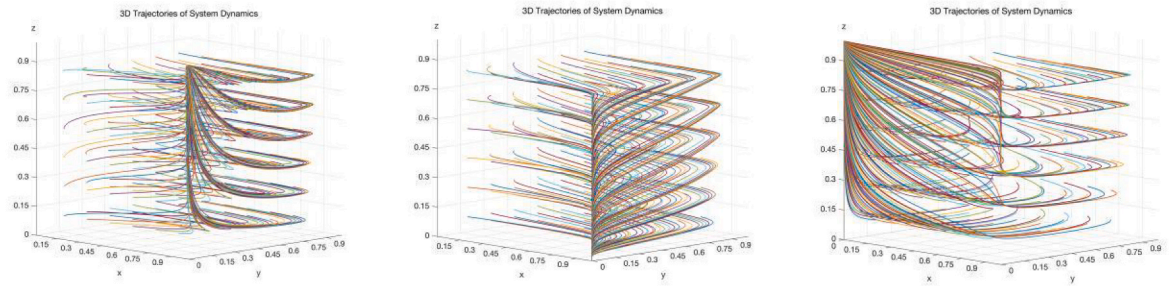
Case 3: Stability of E_7 (0, 0, 1) under specific conditions:

$$F+a\eta w(1+h)>C_B-C_N \tag{4.31}$$

Equilibrium E_7 (0, 0, 1) stabilizes under specified conditions, reflecting a scenario where the e-commerce platform adopts blockchain technology, merchants continue false advertising, and consumers actively defend their rights. The condition $F+a\eta w(1+h)>C_B-C_N$ indicates that platforms' cost of adopting blockchain technology is sufficiently low compared to the benefits derived from penalties on

Table 5
Parameter sets to validate model stability.

Group	a	C_B	C_N	w	F	h	η	G	C_t	C_q	s	σ
E_3 (1, 0, 1)	0.3	100	20	10	20	2	0.7	10	5	2	0.3	0.5
E_4 (1, 0, 0)	0.3	100	20	10	20	2	0.7	3	6	1	0.3	0.5
E_7 (0, 0, 1)	0.3	100	20	60	80	2	0.7	10	5	2	0.3	0.5



(a) (b) (c)

Fig. 3. Evolutionary trajectories of the three parameter sets.

Table 6
Comparison of parameter sets.

Set number	Key parameter set	Dynamic behavior
Fig. 4 (a) Set 1	$C_B = 50$, $C_N = 30$, $F = 10$, $w = 60$, $G = 15$	System exhibits stable transitions initially but loses stability in later stages.
Fig. 4 (a) Set 2	$C_B = 60$, $C_N = 20$, $F = 10$, $w = 130$, $G = 15$	System maintains stability; however, the SEP shifts periodically.
Fig. 4 (a) Set 3	$C_B = 50$, $C_N = 30$, $F = 10$, $w = 85$, $G = 15$	The dynamic path deviates significantly and fails to converge to an ideal equilibrium.
Fig. 4 (a) Set 4	$C_B = 50$, $C_N = 30$, $F = 20$, $w = 130$, $G = 40$	The SEP is properly reached, but system stability is not ensured.
Fig. 4 (b)	$C_B = 50$, $C_N = 30$, $F = 10$, $w = 130$, $G = 20$	System achieves a proper dynamic path with relatively stable performance around the SEP.

merchants and increased sales from consumer trust $a\eta w(1 + h)$. This cost-effectiveness drives the platform to embrace blockchain technology, aligning with consumer interests in defending their rights.

5. Numerical simulation

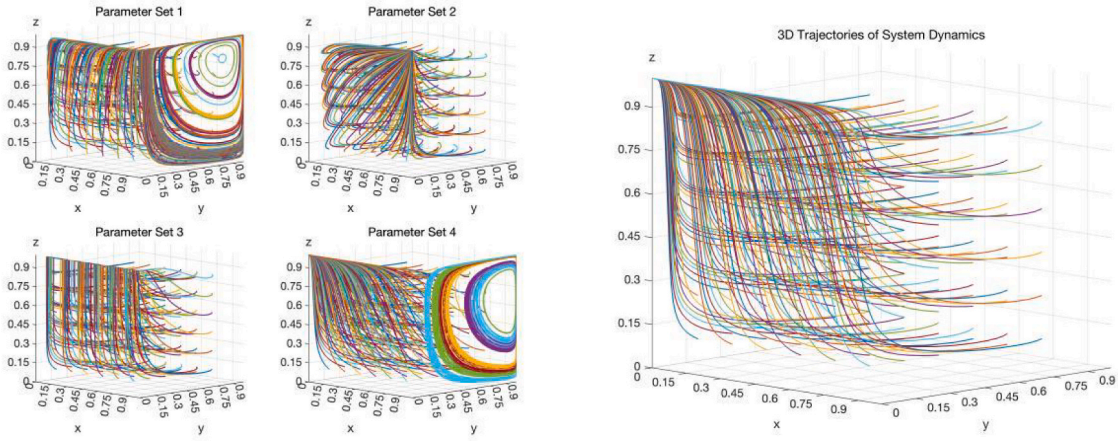
We now validate the theoretical findings through numerical simulation and further investigate the effects of key parameters. This shows how variations in these parameters influence the effectiveness of blockchain technology in curbing false advertising in live e-commerce.

5.1. Verification of stability analysis

First, we conduct a validation analysis of the model. In section 4.4, we established the theoretical framework for analyzing the stability of the evolutionary game model and identified the conditions under which three equilibrium points (E_3, E_4, and E_7) are stable, as defined in Table 4. To validate these findings, we assign suitable parameter sets based on the stability conditions (4.29), (4.30), and (4.31) for simulating the evolutionary dynamics. We ensure that the model assumptions and stability conditions for the equilibrium points are satisfied. Table 5 presents the parameter sets.

Fig. 3 illustrates the evolutionary trajectories for three parameter sets: E_3 (1, 0, 1) in chart (a), E_4 (1, 0, 0) in chart (b), and E_7 (0, 0, 1) in chart (c). In chart (a), the strategies stabilize at (1, 0, 1), where platforms avoid blockchain technology, merchants engage in false advertising, and consumers defend their rights. Chart (b) shows stabilization at (1, 0, 0), indicating that platforms still avoid blockchain technology, merchants continue false advertising, and consumers do not defend their rights. In chart (c), the equilibrium is (0, 0, 1), with platforms adopting blockchain technology, merchants refraining from false advertising, and consumers actively defending their rights.

These results support the theoretical conclusions and demonstrate the impact of different parameter sets on stakeholder strategies in live-streaming e-commerce. Overall, these results confirm the stability conclusions derived from the theoretical analysis, illustrating the influence of different parameter sets on stakeholders' strategic behaviors in live-streaming e-commerce.



(a) (b)

Fig. 4. Evolutionary trajectories of tests.

Table 7
Benchmark array parameter values.

Parameter	Value	Description	Selection Basis
a	0.3	Commission rate on merchant revenue	Derived from literature; typical commission rates range between 2 % and 70 %
C_B	50	Cost for e-commerce platform under blockchain technology	Determined through practical exploration and simulation testing
C_N	30	Cost for e-commerce platform under non-blockchain technology	Determined through practical exploration and simulation testing
w	130	Benchmark revenue for merchants under baseline conditions	Optimized via practical exploration and multiple simulations to ensure stable dynamic behavior
F	10	Penalty for false advertising by brand	Refined through practical exploration to reflect real-world dynamics
h	0.3	Sales growth factor attributable to false advertising	Based on expert interviews and model assumptions, accounting for long-term effects and evolutionary dynamics
η	0.3	Proportionate increase in sales revenue under blockchain technology	Derived from case studies, for example, JD.com's adoption of blockchain technology with an average sales increase of 30 %
G	15	Consumer gain from reporting false advertising	Adjusted through simulation to match favorable evolutionary trajectories
C_t	8	Cost of consumers defending their rights under non-blockchain technology	Set based on literature and expert opinions to reflect higher costs without blockchain technology
C_q	5	Cost of consumers defending their rights under blockchain technology	Set based on literature and expert opinions to reflect higher costs without blockchain technology
s	0.3	Proportionate increase in consumer awareness of defending their rights due to blockchain technology	Derived from literature on RFID technology and its impact on consumer trust and behavior
σ	0.5	Intensity of stochastic disturbance	Referenced from literature to capture random fluctuations in dynamic processes

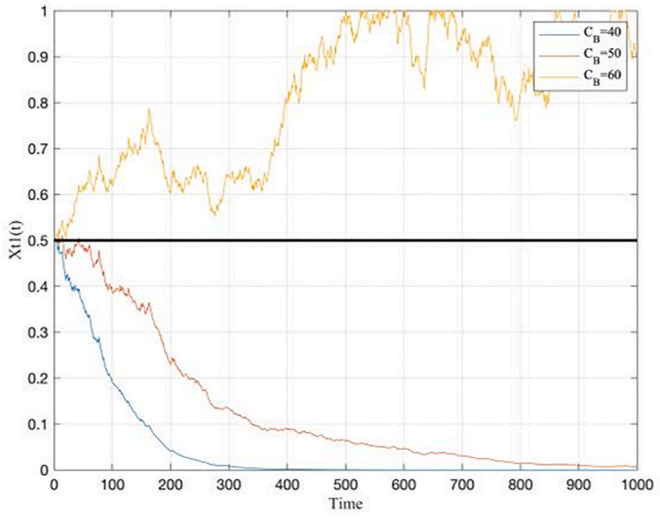


Fig. 5. Influence of parameter C_B on the choice of E-commerce platform strategy.

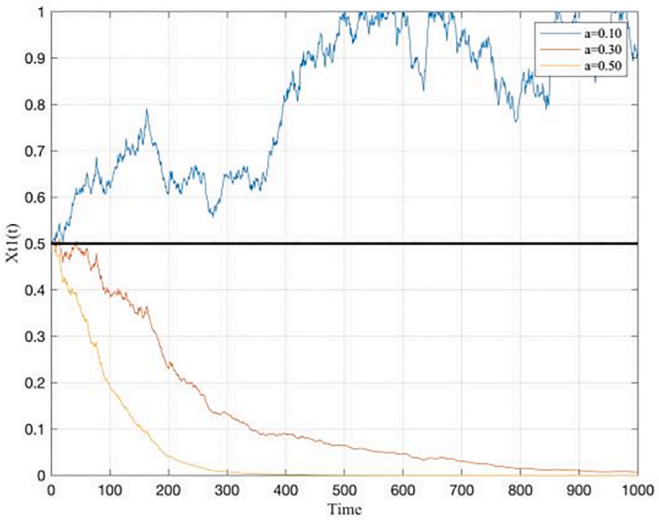


Fig. 6. Effect of parameter a on the choice of E-commerce platform strategy.

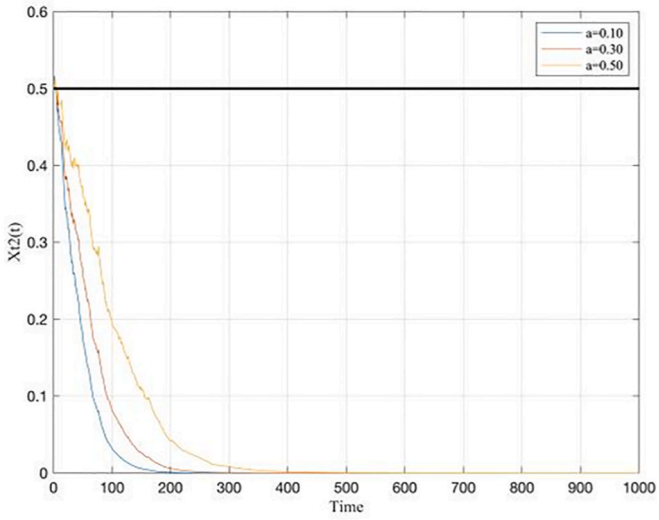


Fig. 7. Impact of parameter a on merchants' strategy choice.

5.2. Basis for parameter selection

This study investigates how blockchain technology mitigates false advertising in live-streaming e-commerce. Therefore, we emphasize the equilibrium point E_7 (0, 0, 1), where e-commerce platforms adopt blockchain technology, merchants engage in false advertising, and consumers choose to defend their rights.

Although some live e-commerce enterprises have adopted blockchain technology, challenges such as corporate privacy protections and the complexity of stochastic models make it difficult to obtain suitable first-hand data from actual applications to define parameter settings. Consequently, the selection of parameter sets needs refinement. The benchmark parameter set for the sensitivity analysis was determined based on a literature review and practical exploration, ensuring that the model and parameter assumptions reflect realistic dynamics in live-streaming e-commerce.

(1) Literature review

To establish the refined parameter range, we conducted an extensive literature review and analyzed real-world cases, focusing on research in related industries. For example, a critical

feature of blockchain technology is enhancing supply chain transparency to increase consumer trust. To this end, we studied application data from radio frequency identification (RFID) technology, which is widely applied and shares similar functionality. We concluded that consumers are more likely to defend their rights under traceable technologies (Aiello et al., 2015; Dong et al., 2023; Piramuthu et al., 2013; Yao and Zhu, 2020). Based on this, we set parameter s to 0.3, ensuring that the cost of consumers defending their rights under blockchain technology ($C_q = 5$) remains lower than the cost under non-blockchain technology ($C_t = 8$).

Regarding the commission rate factor, commissions typically range between 2 % and 70 %; thus, we set parameter a to 0.3 (Cui et al., 2024). To capture the proportionate increase in sales revenue associated with blockchain technology, we also considered case studies like JD.com, which reported a 30 % increase in sales for stores that adopted blockchain technology. Based on JD.com's published reports (JD Digits, 2020), this finding guided the selection of parameter η, which was also set to 0.3.

(2) Practical exploration

(i) Expert interview

In addition to the literature review, we conducted an in-depth interview with an advertising expert from a well-known e-commerce platform in China. The expert, a designer at the e-commerce platform, mentioned that the initial goal of most advertising campaigns is to achieve a 20–40 % increase in sales. However, in certain short-term, high-intensity campaigns (e.g., big promotional events), the returns can temporarily be as high as 200–300 %. He emphasized that such short-term gains are often accompanied by higher costs and significant long-tail effects. Therefore, when considering the long-term impact of false advertising on sales growth, we adopted a more conservative estimate. Based on the expert's recommendations and model assumptions, sales growth coefficient h was set to 0.3, reflecting the evolutionary dynamics over a longer horizon.

(ii) Parameter refinement and testing

In section 5.2 (1), key parameters (e.g., s, C_q, C_t, a, η, h) were theoretically established and validated based on a literature review and case studies. However, for certain parameters (e.g., C_B, C_N, w, F, G), their dynamic behaviors in different scenarios were challenging to determine solely through literature. Thus, we refined these parameters via practical exploration and multiple simulation tests.

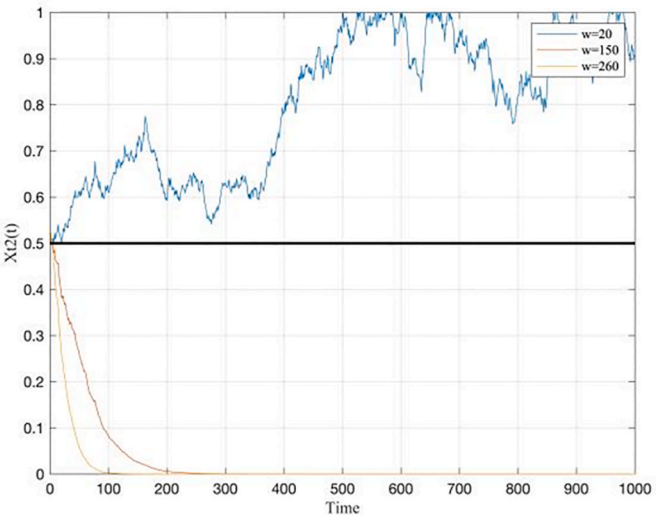


Fig. 9. Effect of Parameter w on Merchants' Strategy Choice.

In section 5.1, Fig. 3(c) showed that the equilibrium point eventually stabilized at (0, 0, 1), confirming the rationality of the parameter set. However, the evolutionary process appeared coarse due to the lack of precision in parameter selection. To address this, we conducted additional tests to refine parameters and assess the robustness of the model under extreme conditions, enhancing the reliability of the findings. Through repeated testing and adjustments (Table 6), we derived an optimized parameter set that resulted in a more favorable evolutionary trajectory shown in Fig. 4 (b). Hence, we present the final optimized parameter set in Table 7, which demonstrates favorable trends and high fitness, providing a better set for the sensitivity analysis.

Table 7 summarizes the final parameter values and their interpretations and sources, which presents the benchmark array of parameter values used for further analysis.

By combining an extensive literature review, expert insights, and simulation testing, we ensure that the parameter settings are both theoretically grounded and empirically supported, thereby enhancing the reliability and credibility of the findings. Although real-world data on false advertising in live-streaming e-commerce is limited due to privacy

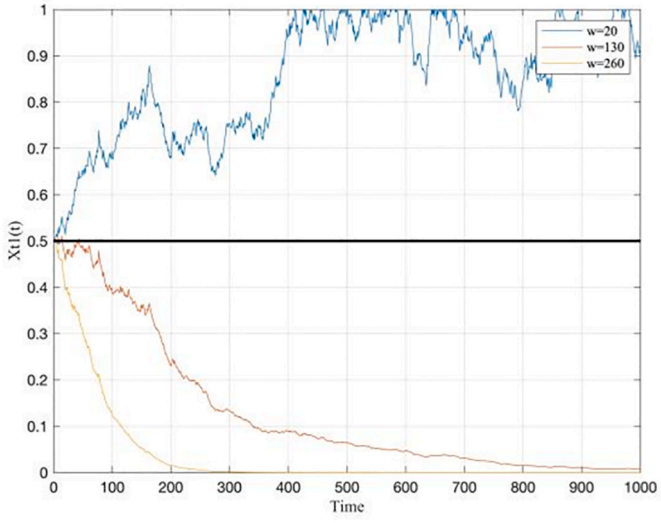


Fig. 8. Effect of Parameter w on the Choice of E-commerce Platform Strategy.

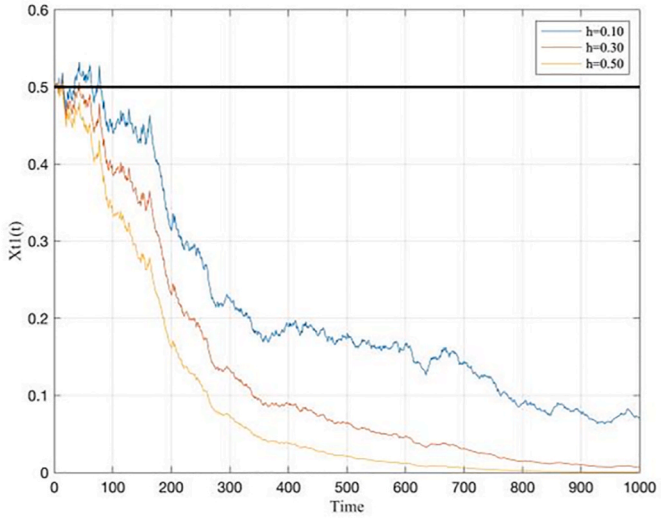


Fig. 10. Effect of Parameter h on the Choice of E-commerce Platform Strategy.

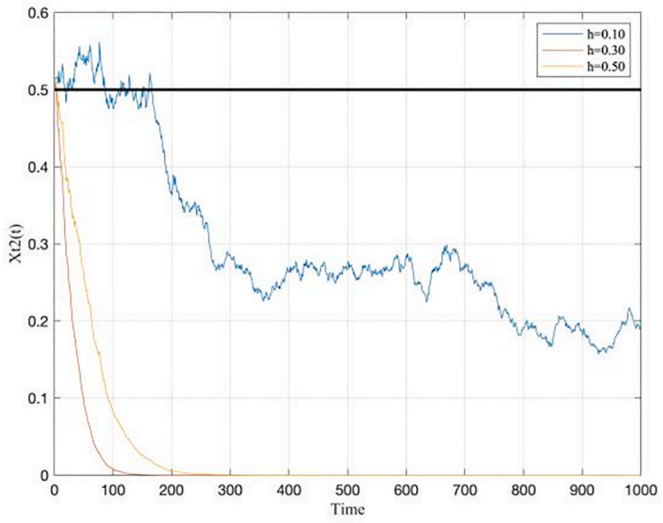


Fig. 11. Effect of Parameter h on Merchants' Strategy Choice.

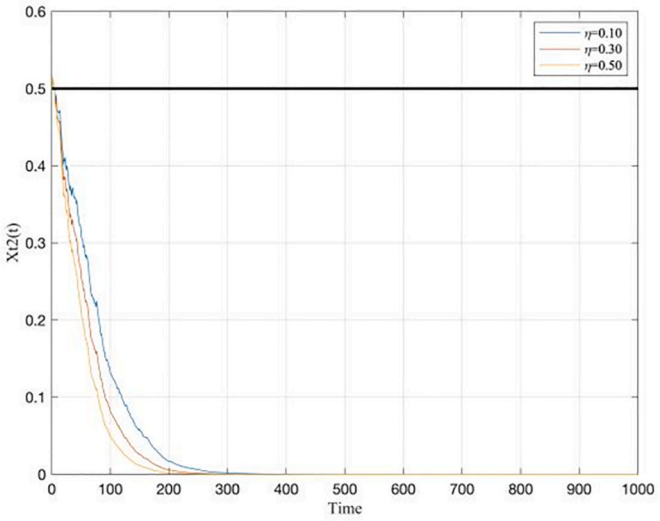


Fig. 13. Impact of Parameter η on Merchants' Strategy Choice.

constraints, the application of blockchain technology in case studies like JD.com, along with expert testimony and empirical simulations, provides a robust foundation for our model. Future research could further validate these findings by incorporating additional real-world case data.

5.3. Sensitivity analysis of key factors

To clarify the impact of key parameters on the strategy choices of e-commerce platforms, merchants, and consumers, we conducted separate strategy choice analyses for each participant. This approach enabled deconstructing the key factors from various perspectives. We also made both cross-sectional and longitudinal comparisons to observe how the same parameter variation affects the strategies of different participants. This analysis shows how these parameters influence participants' decision-making processes in the evolutionary game. In the sensitivity analysis, the x-axis represents the number of iterations (1000 times) and y-axis the probability of strategy selection, ranging from 0 to 1.

(1) Analysis of economic factors

(i) E-commerce platform costs under blockchain technology

Fig. 5 shows the impact of e-commerce platform cost C_B on e-commerce platform strategy choice under blockchain technology. It clarifies that the higher the cost of blockchain technology to the platform, the more reluctant the platform is to proceed with the adoption thereof. In other words, elevated costs act as a barrier to blockchain technology adoption, underscoring the necessity of meticulous cost-benefit analyses when selecting and implementing technology.

(ii) Commission coefficient

Figs. 6 and 7 show the effect of commission coefficient a on the strategy choices of e-commerce platforms and merchants. At different values of the commission coefficient, there is a significant difference in the strategy choices of e-commerce platforms and merchants.

For e-commerce platforms, the probability of choosing to adopt blockchain technology in the pre-evolutionary period increases significantly as the commission coefficient increases. This suggests that in a high commission environment, e-commerce platforms are more willing to invest in blockchain technology to improve transaction transparency, reduce the cost of false advertising, and enhance the credibility of the platform. However, in the case of lower

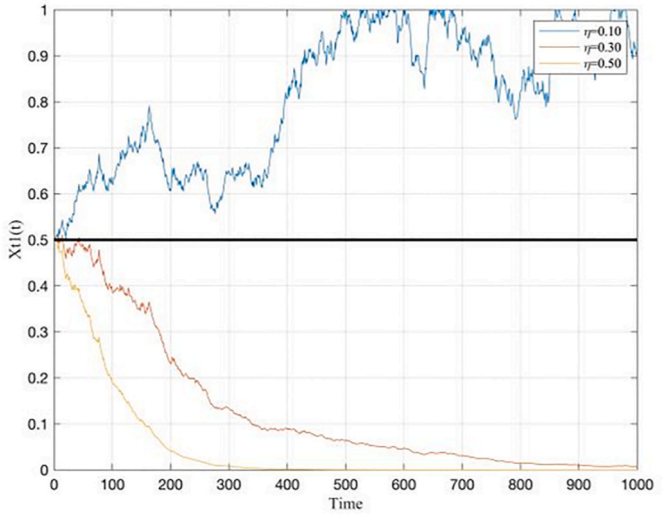


Fig. 12. Effect of Parameter η on the Choice of E-commerce Platform Strategy.

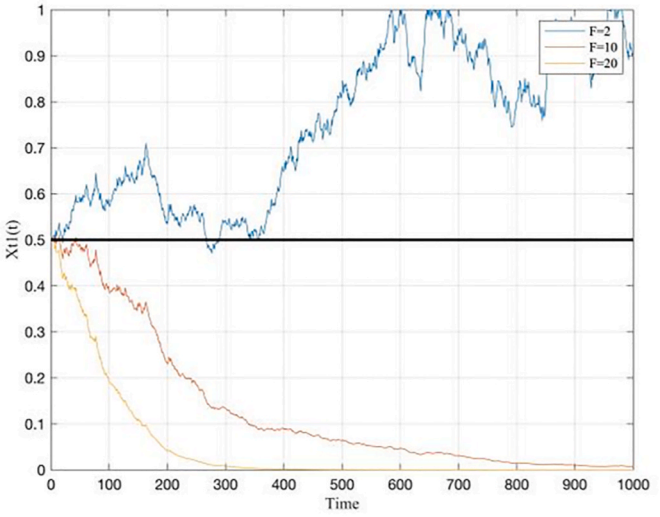


Fig. 14. Effect of parameter F on the choice of E-commerce platform strategy.

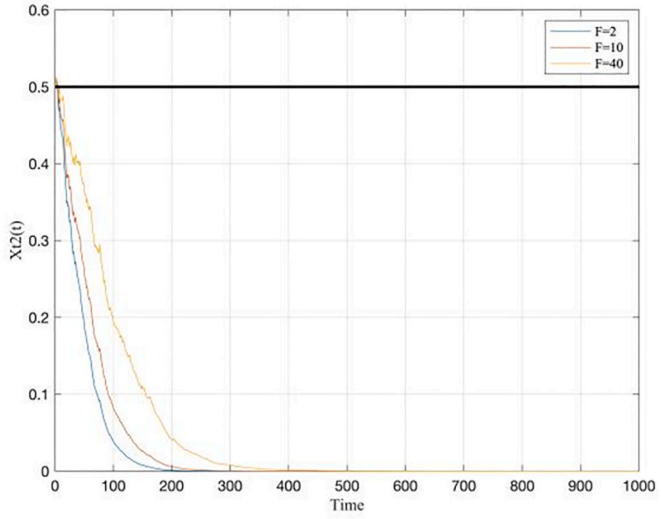


Fig. 15. Effect of parameter F on merchants' strategy choice.

commission coefficients, e-commerce platforms are more likely to choose not to adopt blockchain technology, especially when commission coefficients are low ($a = 0.1$) and their evolutionary strategies tend to not use blockchain technology. This reflects the sensitivity of commission revenue to the technology choice of e-commerce platforms.

For merchants, a rise in the commission coefficient increases the probability of e-commerce platforms adopting blockchain technology, increases transaction transparency, increases the probability of merchants adopting truthful advertising, and decreases the probability of choosing to engage in false advertising. However, the magnitude of this change is small compared to that for e-commerce platforms and ultimately tends toward false advertising. Merchants may be feeling more pressure on advertising costs in a low-commission environment, and are therefore more inclined to use false advertising to reduce actual product quality and drive sales. In addition, lower commissions prompt merchants to perceive less responsibility on the part of the platform, a lower probability of being actively regulated by the platform, and a higher probability of false advertising. In summary, with high commissions, e-commerce platforms have increased revenue linkages with merchants, and therefore rely more on building a good brand reputation to attract consumers and drive merchants to focus on long-term branding. Low commissions may lead merchants to favor short-term false advertising strategies.

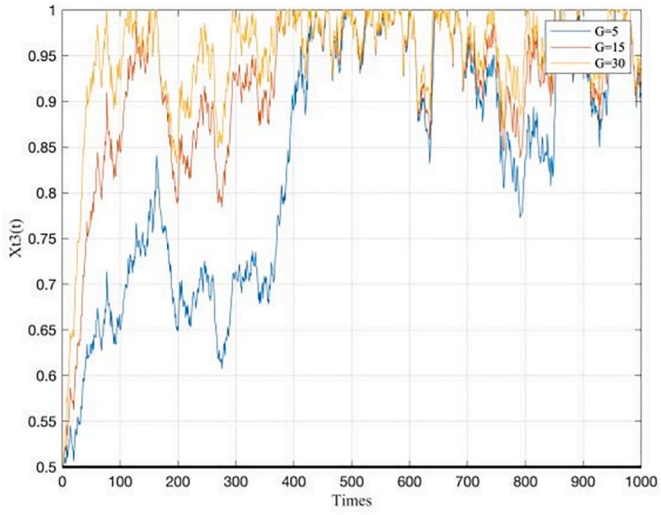


Fig. 16. Effect of parameter G on consumer strategy choice.

(iii) Merchants' revenue

Figs. 8 and 9 show the impact of merchants revenue w on the e-commerce platform and merchants' strategy choice. As shown, when $w = 150$ and $w = 260$, the evolution trajectories of both are similar. However, when $w = 20$, which means brand merchants' revenue drops significantly, the evolution strategies of e-commerce platforms and brand merchants differ significantly.

When merchants' revenue rises, e-commerce platforms are more willing to adopt blockchain technology to improve their credibility and trustworthiness and attract more merchants and consumers. However, in low revenue scenarios, e-commerce platforms may focus more on cost effectiveness and choose non-blockchain technology.

In high-revenue environments, merchants are more inclined to adopt false advertising and seek to maximize sales revenue in the short term. This may stem from short-termist business strategies brought about by high revenue levels, whereas the positive incentives of revenue drive merchants to focus on maximizing profits quickly, which leads them to neglect long-term brand reputation and sustainable operations. In addition, high earnings make merchants more confident in their market position, believing that they can stay ahead in a competitive market even if they adopt false advertising and that their earnings can easily offset the pressure from fines imposed by platforms and cost of consumers' rights. This makes them more willing to risk false advertising. Conversely, a trend toward more cautious behavior emerges when merchants' revenues are lower. Merchants focus more on building long-term brand reputation in low-revenue environments, opting for more truthful promotions to ensure lasting consumer trust. This suggests that merchants are more inclined to develop robust promotional strategies and focus on long-term relationships with consumers when facing economic pressures.

Therefore, e-commerce platforms should give graded supervision and treatment to merchants of different sizes and revenues to achieve the desired effect of combating false advertising.

(iv) False advertising benefit

Figs. 10 and 11 show the impact of false advertising benefit factor h on the strategy choices of e-commerce platform and brand owners. As shown, the change of the false advertising benefit factor significantly impacts merchants more than e-commerce platforms.

As the false advertising benefit factor increases, e-commerce platforms are more inclined to rapidly adopt blockchain technology. This is due to their high sensitivity to the risk of false advertising, where efficient blockchain technology is significantly advantageous in reducing the risk. As the false advertising benefit factor increases, e-commerce platforms are more willing to adopt secure and transparent technology to maintain their reputation and attract merchants and consumers.

For merchants, an increase in the false advertising benefit factor drives a greater tendency to adopt false advertising. However, when the false advertising benefit factor decreases to 0.1, merchants' willingness to adopt false advertising decreases significantly. This suggests they are sensitive to the benefits of false advertising, and when the benefit factor is lower, they are more inclined to choose more truthful and

credible propaganda.

This requires that consumers improve their ability to identify false promotions to achieve more rational consumer decisions, which helps regulate merchants' behavior.

(2) Analysis of technological factors

Figs. 12 and 13 show the impact of blockchain technology benefit factor η on the strategy choices of e-commerce platforms and merchants. As shown, the change in the false advertising benefit factor significantly affects e-commerce platforms more than merchants.

For e-commerce platforms, the benefits of blockchain technology reflect consumers' recognition and acceptance of the technology and their aversion to false advertising. High blockchain technology benefits motivate e-commerce platforms to prefer it for better reputation and revenue security. In particular, at $\eta = 0.1$, blockchain technology benefits are low and e-commerce platforms may choose not to adopt it to maximize benefits. This suggests that blockchain technology is not only a technology choice for e-commerce platforms, but also a strategic decision to balance credibility and revenue.

For merchants, the impact of blockchain technology benefits on their choice of promotional methods is relatively weak. As blockchain technology benefits rise, the probability of them choosing false publicity slightly increases. This may reflect merchants' relative indifference to the actual impact of blockchain technology in terms of advertising, which combined with Fig. 11, suggests that consumer perceptions of and reactions to publicity more significantly impact their choices.

(3) Analysis of regulatory factors

Figs. 14 and 15 show the impact of e-commerce platform fines F on the strategy choices of e-commerce platforms and merchants. As shown, the change in e-commerce platform fines significantly affects e-commerce platforms more than merchants.

For e-commerce platforms, fines are seen as a defense of consumer rights and regulation of malpractices in the e-commerce market. When fines are low, that is, the penalty for false advertising for merchants is small, the impact is weak, especially at $F = 2$, and platforms prefer blockchain technology to fines. In contrast, when the fines are higher, the impact on the brand owner becomes larger, and leads to higher enforcement costs for the e-commerce platform. Gradually, e-commerce platforms then tend to choose blockchain technology and stabilize on adopting it.

For brand owners, the increase in fines has curbed merchants' false advertising behavior, although merchants still choose to continue false advertising. This shows that when merchants weigh the cost of fines against the benefits of false advertising, false advertising is more important for sales and market share acquisition.

(4) Analysis of behavioral factors

Fig. 16 shows the impact of consumer reporting of false advertising gains G on consumer choice. It shows that the probability of consumers choosing to defend their rights tends to increase as the gain from reporting increases. This underscores the strong relationship between consumer reporting behavior and their awareness of rights protection. It also suggests that increasing reporting incentives can effectively reduce false advertising in the market by encouraging proactive consumer participation.

6. Discussion

6.1. Theoretical contributions

(1) Extension of stochastic evolutionary game theory to live-streaming e-commerce ecosystems under blockchain technology

Our study advances the application of game theory by introducing a stochastic evolutionary game framework tailored to live-streaming e-commerce ecosystems under blockchain technology. By modeling strategic interactions among e-commerce platforms, merchants, and consumers, we identify how economic factors (e.g., revenue sensitivity, cost structures, and commission coefficients), technological factors, and regulatory behavioral dynamics influence the effectiveness of blockchain technology in mitigating false advertising. Unlike traditional game-theoretic applications in e-commerce (Agrawal et al., 2023; Song et al., 2023; Wang J. et al., 2021), our model integrates stochastic disturbances to account for the dynamic and uncertain interactions within the ecosystem. This approach offers a deeper understanding of the strategic behaviors driving technology adoption and false advertising mitigation in digital marketplaces.

(2) Advancement of information asymmetry theory in e-commerce

Our study examines how revenue and commission structures influence the adoption of blockchain technology by e-commerce platforms. Our findings show that higher commission rates incentivize platforms to adopt blockchain technology to enhance credibility, while merchants respond by adjusting their advertising strategies, balancing between truthful and false promotions. This aligns with previous research (Tan and Salo, 2023) and highlights the role of blockchain technology in reducing false advertising. Our work underscores the importance of using technology to build trust and transparency in the e-commerce ecosystem.

(3) Integration of behavioral economics and technology adoption theory

Technology adoption is a long-term process shaped by diverse factors and stakeholders' interests (Zhou et al., 2020, 2024). In live-streaming e-commerce, recent studies reveal a strong link between commission structures and strategic behaviors among stakeholders, underscoring the complexity of integrating innovations like blockchain (Zhu et al., 2024; Dai and Zhang, 2025). Building on this, our research further integrates behavioral economics with technology adoption theory to examine how economic incentives and behavioral factors shape the decision-making processes of platforms and merchants. We find that platforms are more likely to adopt blockchain technology in high-commission environments, where the incentive to enhance transparency and reduce the cost of false advertising is stronger. In contrast, merchants' behaviors are influenced by their revenue levels—high-revenue contexts tend to drive merchants toward short-term gains, often through false advertising, while low-revenue environments encourage long-term branding and truthful promotions. These insights help platforms and merchants make more informed decisions when adopting blockchain technology, balancing short-term profitability with long-term brand integrity.

(4) Collaborative governance framework for addressing false advertising in digital ecosystems

Previous studies have shown that while blockchain technology enhances transparency, its effectiveness in mitigating deceptive practices relies on the combined efforts of platforms, merchants, and consumers (Liu et al., 2024; Xu et al., 2025). Our study further reinforces and validates the necessity of multi-stakeholder collaboration in addressing false advertising in e-commerce ecosystems. This collaborative approach ensures more robust enforcement and helps create a sustainable digital marketplace. Our framework offers insights into how regulatory strategies can integrate technological tools like blockchain technology and collaborative governance to reduce false advertising and foster industry-wide growth.

6.2. Managerial implications

(1) Strategic decision-making for e-commerce platforms and merchants

Our findings suggest practical strategies for e-commerce platforms to balance technological investment with profitability. Platforms can selectively adopt blockchain technology, focusing on high-revenue merchants where enhancing credibility aligns with revenue growth. Initially, platforms should target top streamers and leading merchants to maximize impact, setting industry benchmarks and combating false advertising at a more significant scale.

(2) Revenue-based regulation for e-commerce platforms

Platforms should adopt a graded supervision model based on merchants' revenue levels. For example, high-revenue merchants better able to absorb costs should face stricter monitoring to deter false advertising, while low-revenue merchants could benefit from support programs that incentivize sustainable brand-building practices, ensuring long-term business growth.

(3) Cost management and blockchain technology adoption

Our study highlights the importance of cost considerations in adopting blockchain technology. Platforms can collaborate with merchants to share the implementation costs, which can help foster the creation of transparent quality traceability systems. In addition, integrating blockchain technology with existing platform technologies offers a more cost-effective way to streamline operations and reduce development expenses.

(4) Enhancing platform-merchant collaboration

Our findings show that high commission coefficients strengthen the alignment between platform and merchant revenues, encouraging collaborative investment in blockchain technology. Platforms should establish incentive mechanisms that promote truthful advertising practices and align merchant strategies with long-term brand reputation goals.

(5) Emphasizing consumer engagement

Platforms and merchants should actively encourage consumers to value their feedback as a tool for shaping market credibility. Leveraging blockchain technology to ensure the authenticity and visibility of genuine reviews motivates consumers to view their evaluations as a means of protecting their long-term interests, thereby fostering market accountability.

6.3. Broader implications

This study contributes to the growing body of research on digital ecosystem sustainability, highlighting the pivotal role of technology in shaping platform and merchant behavior. The findings underscore the importance of cost-effective technology adoption and adaptive regulation in balancing economic and ethical objectives in the digital marketplace (Cano et al., 2023; Xin et al., 2025). Specifically, blockchain technology, alongside other emerging technologies such as artificial intelligence, offers significant opportunities to address challenges posed by new business models. While these technologies can enhance transparency, trust, and efficiency, their adoption remains a long-term decision shaped by various internal and external factors including market readiness, regulatory environments, and technological capabilities (Dehghani et al., 2022; Kumar and Taneja, 2024).

Our research seeks to clarify the complex challenges in the live-streaming e-commerce landscape, offering insights that can guide more informed decision-making. By focusing on the interaction between technology, regulation, and platform strategy, we present a practical framework for platforms and merchants to adopt blockchain technology in ways that enhance both credibility and profitability. Although strategic decisions are influenced by various factors, they are more effective when contextualized within specific scenarios. This approach empowers stakeholders to make better choices and maximize the impact of their technological investments. Furthermore, the framework presented here can serve as a valuable resource for other scholars studying the adoption of new technologies, such as artificial intelligence.

In conclusion, our study underscores the importance of aligning economic strategies with technological and regulatory advancements. The theoretical and practical contributions outlined in this work provide a solid foundation for shaping the future of e-commerce ecosystems under blockchain-enabled transparency. These insights offer useful guidance for both academics and practitioners navigating the complex intersection of technology, market forces, and regulation in digital commerce.

7. Conclusion

7.1. Key findings

This study identifies several key findings regarding blockchain technology adoption in live-streaming e-commerce, particularly in mitigating false advertising:

(1) High commission rates align the interests of platforms and merchants, encouraging merchants to adopt truthful advertising strategies and fostering a more transparent marketplace.
(2) In high-profit environments, the temptation for merchants to engage in false advertising increases. This drives platforms to adopt blockchain technology to enhance transparency and counterbalance short-term profit-driven behaviors.
(3) Blockchain technology adoption is more likely when its long-term benefits outweigh initial costs. While it improves platform reputation and credibility, its direct impact on reducing false advertising by merchants is more limited.
(4) Regulatory measures such as fines for false advertising and consumer reporting rewards significantly increase the likelihood of blockchain technology adoption. These measures raise the cost of deceptive practices and support the creation of a more transparent and trustworthy marketplace.

7.2. Practical implications

(1) For e-commerce platforms

Platforms can leverage blockchain technology to improve transparency and align merchant behavior with long-term reputation goals, especially in high-commission environments. They should also consider implementing regulatory measures (e.g., fines, reporting rewards) to complement blockchain technology adoption and further enhance credibility.

(2) For merchants

Merchants should be encouraged to adopt truthful advertising strategies to avoid penalties and benefit from the long-term advantages of blockchain technology, such as enhanced consumer trust and sustainable brand-building.

(3) For consumers

Consumers can use their influence to foster transparency by actively

engaging in reporting and providing feedback on merchant practices. This active participation can further promote truthful advertising and a more reliable e-commerce environment.

(4) For policymakers and regulators

Governments should focus on refining legal frameworks related to false advertising, ensuring clear delineation of responsibilities for all involved parties. To facilitate blockchain technology adoption, policymakers should consider offering tax incentives for small and medium-sized blockchain technology companies, thus encouraging innovation and development. Additionally, supporting the establishment of blockchain-based evidence platforms and promoting the use of mature blockchain technologies will further strengthen the integrity and traceability of e-commerce activities, ultimately enhancing consumer protection.

7.3. Limitations and future work

7.3.1. Research limitations

This study provides valuable insights but has several limitations. First, it does not fully consider individual consumer differences such as varying technological acceptance levels or behavioral traits, which could influence the adoption and effectiveness of blockchain technology. Second, the model focuses on the strategic interactions among e-commerce platforms, merchants, and consumers, without considering external factors like macroeconomic conditions or changing consumer behaviors. It assumes uniform revenue models and commission structures across platforms, which may not reflect the diversity of real-world e-commerce environments. Finally, while the study offers a theoretical analysis of blockchain technology adoption, empirical validation in real-world settings remains a crucial area for future research.

7.3.2. Future research directions

Future research could explore several important areas to expand on the findings of this study. First, cross-cultural comparisons could assess how blockchain adoption and false advertising mitigation vary across regions, offering insights for global platforms. Second, long-term effectiveness studies are needed to explore the sustained impact of blockchain technology on consumer trust and platform strategies over time. Third, alternative technological solutions such as decentralized finance or machine learning should be examined for their potential to complement or outperform blockchain technology in some aspects of e-commerce. Fourth, the model could be extended to include additional stakeholders such as third-party auditors, governmental bodies, or consumer advocacy groups to capture a more holistic view of the ecosystem and their roles in mitigating false advertising. Finally, diverse types of false advertising such as exaggerated claims, omission of critical information, or manipulative endorsements should be explored to evaluate the differential effectiveness of blockchain technology in addressing these issues.

CRediT authorship contribution statement

Ruipeng Wang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuhong Tai:** Supervision, Funding acquisition.

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Data will be made available on request.

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